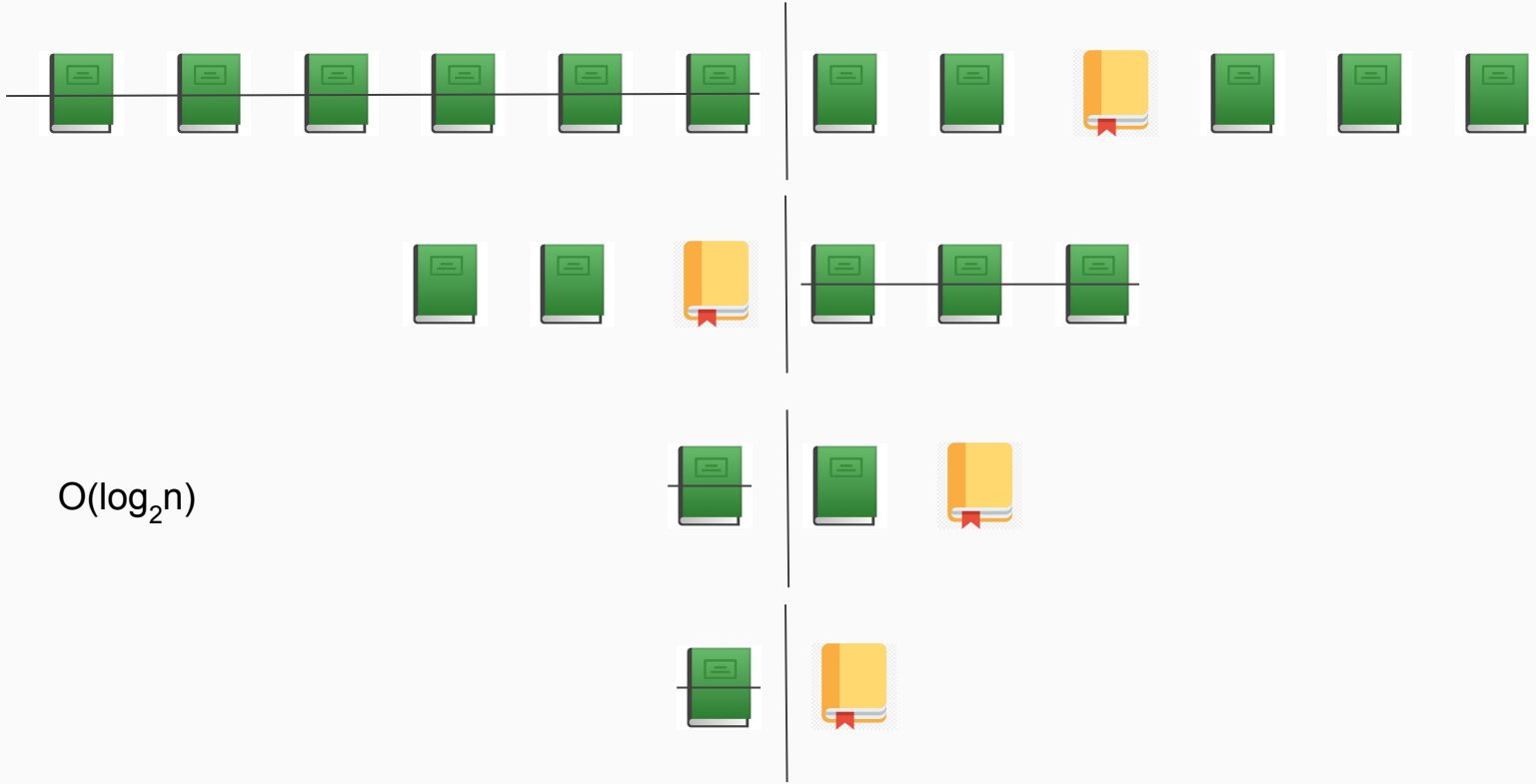
# Bisection method 二分法

BinarySearch 二分查找

代码魔术师
magiciendecode.fr

# Bisection 二分法



$O(\log_2 n)$

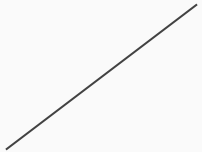


代码魔术师
magiciendecode.fr

# BinarySearch 二分查找

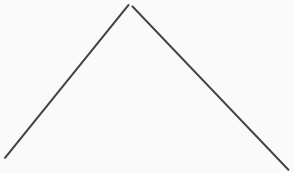
- Given collection has a regular sequence
- Find a position that satisfy some conditions

-1, 2, 5, 7, 11, 100, 150 ...



1, 1, 1, 1, 1, 0, 0, 0, 0

-9, -5, 1, 8, 17, 6, 5, 2, 0, -3, -10





代码魔术师
magiciendecode.fr

# When and How?

When: search an index or an element in a regular range
Time complexity: log (n)

Template: https://github.com/MagicienDeCode/LeetCode_Solution/blob/master/src/BinarySearch.md

https://github.com/MagicienDeCode/py3_interview/blob/master/basic_grammar/binary_search/704binary_search_answer.py

⇒ left right ?
⇒ while( left + 1 < right ) // avoid dead loop
⇒ conditions
⇒ right first or left first ?
⇒ left = mid / right = mid

1. search the ( exact ) ( position / element )
2. search the ( first / last ) ( position / element )
3. search with complex conditions
4. binary search the result range with a verification function

代码魔术师
magiciendecode.fr

# 1. search the ( exact ) ( position / element )

704. Binary Search (exact element)
167. Two Sum II - Input array is sorted (exact element)
35. Search Insert Position ( [first index] <= target)
374. Guess Number Higher or Lower
392. Is Subsequence (follow up using a hashmap to store index of char, then binary search in list)
852. Peak Index in a Mountain Array
162. Find Peak Element
74. Search a 2D Matrix (binary search x 2)
240. Search a 2D Matrix II (foreach then search)
475. Heaters

代码魔术师
magiciendecode.fr

# 2. search the ( first / last ) ( position / element )

69. Sqrt(x) (last element ^2 <= target)
34. Find First and Last Position of Element in Sorted Array
278. First Bad Version
350. Intersection of Two Arrays II (first & last then count the total)
1337. The K Weakest Rows in a Matrix (first or last then count)
1351. Count Negative Numbers in a Sorted Matrix
300. Longest Increasing Subsequence (LIS)
441. Arranging Coins
454. 4Sum II

528. Random Pick with Weight (transform then binary search)
497. Random Point in Non-overlapping Rectangles
911. Online Election
981. Time Based Key-Value Store

315. Count of Smaller Numbers After Self
327. Count of Range Sum
354. Russian Doll Envelopes (LIS)
862. Shortest Subarray with Sum at Least K (Deque)
887. Super Egg Drop (Math, also DP)

代码魔术师
magiciendecode.fr

# 3. search with complex conditions

33. Search in Rotated Sorted Array
81. Search in Rotated Sorted Array II
153. Find Minimum in Rotated Sorted Array & 154
275. H-Index II
658. Find K Closest Elements

710. Random Pick with Blacklist

代码魔术师
magiciendecode.fr

# 4. binary search the result range with a verification function

378. Kth Smallest Element in a Sorted Matrix
718. Maximum Length of Repeated Subarray [DP]

875. Koko Eating Bananas
1011. Capacity To Ship Packages Within D Days
1482. Minimum Number of Days to Make m Bouquets

1201. Ugly Number III
1283. Find the Smallest Divisor Given a Threshold
1300. Sum of Mutated Array Closest to Target
1292. Maximum Side Length of a Square with Sum Less than or Equal to Threshold
540. Single Element in a Sorted Array

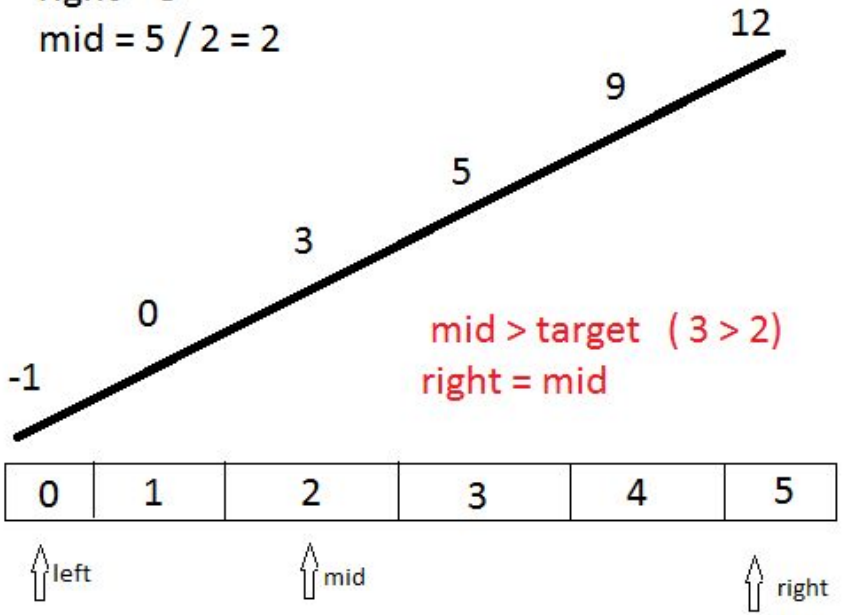
410. Split Array Largest Sum
668. Kth Smallest Number in Multiplication Table
778. Swim in Rising Water (DFS)
1044. Longest Duplicate Substring (Rabin-Karp)

代码魔术师
magiciendecode.fr

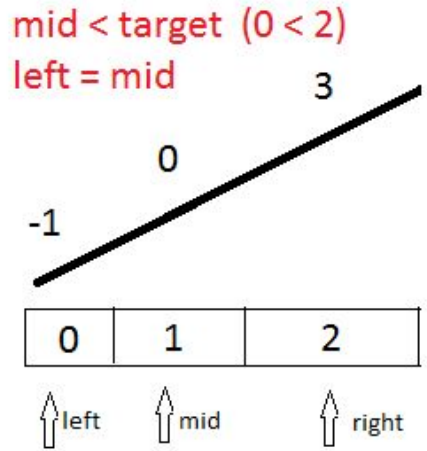
https://leetcode.com/problems/binary-search/

target = 2

left = 0
right = 5
mid = 5 / 2 = 2

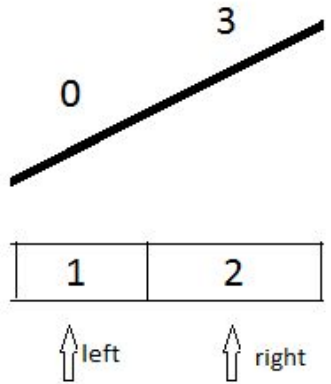



left = 0
right = 2
mid = 1




left = 1
right = 2

while( left +1 < right )





代码魔术师
magiciendecode.fr

# 34. Find First and Last Position of Element in Sorted Array

```
fun searchFirstPosition(nums: IntArray, target: Int): Int {
    if (nums.isEmpty()) {
        return -1
    }
    var left = 0
    var right = nums.size - 1
    while (left + 1 < right) {
        val mid = left + (right - left) / 2
        when {
            nums[mid] == target -> right = mid
            nums[mid] < target -> left = mid
            nums[mid] > target -> right = mid
        }
    }
    return when (target) {
        nums[left] -> left
        nums[right] -> right
        else -> -1
    }
}
```

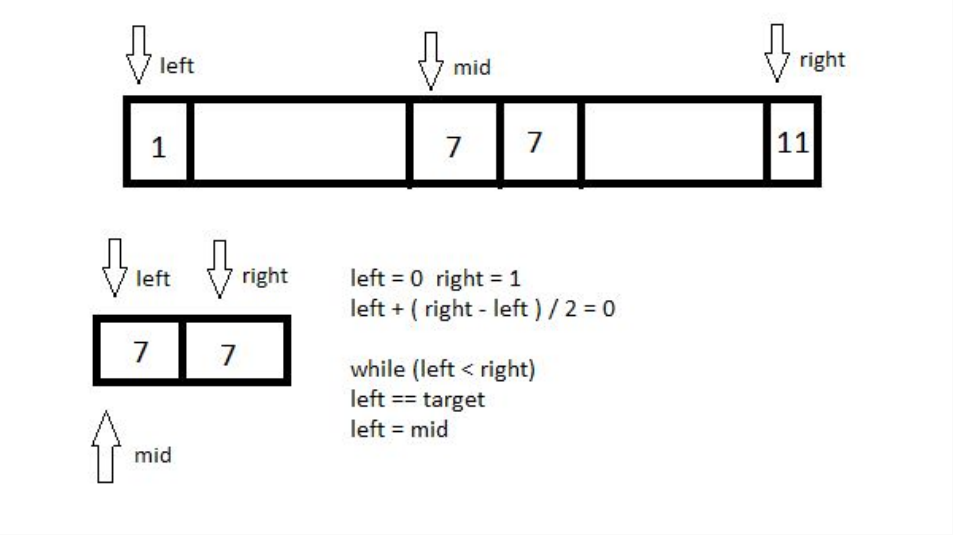
```
fun searchLastPosition(nums: IntArray, target: Int): Int {
    if (nums.isEmpty()) {
        return -1
    }
    var left = 0
    var right = nums.size - 1
    while (left + 1 < right) {
        val mid = left + (right - left) / 2
        when {
            nums[mid] == target -> left = mid
            nums[mid] < target -> left = mid
            nums[mid] > target -> right = mid
        }
    }
    return when (target) {
        nums[right] -> right
        nums[left] -> left
        else -> -1
    }
}
```

代码魔术师
magiciendecode.fr

# Template Java / Kotlin

1. while ( left +1 < right )
2. mid = left + ( right - left ) / 2
3. mid == target ? mid > target ? mid < target
   left = mid or left = mid +1
4. left? right?

Int.MAX_VALUE = 2147483647



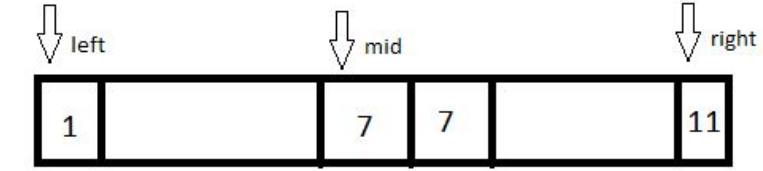

https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/

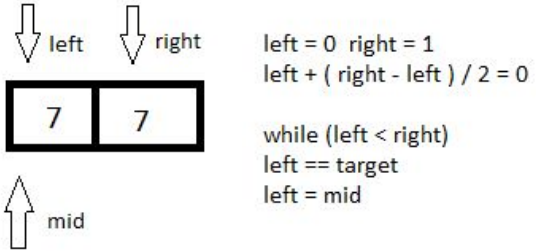
https://leetcode.com/tag/binary-search/


代码魔术师 magiciendecode.fr

# 34. Find first and last position

https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/




left = 0 right = 1
left + ( right - left ) / 2 = 0

while (left < right)
left == target
left = mid

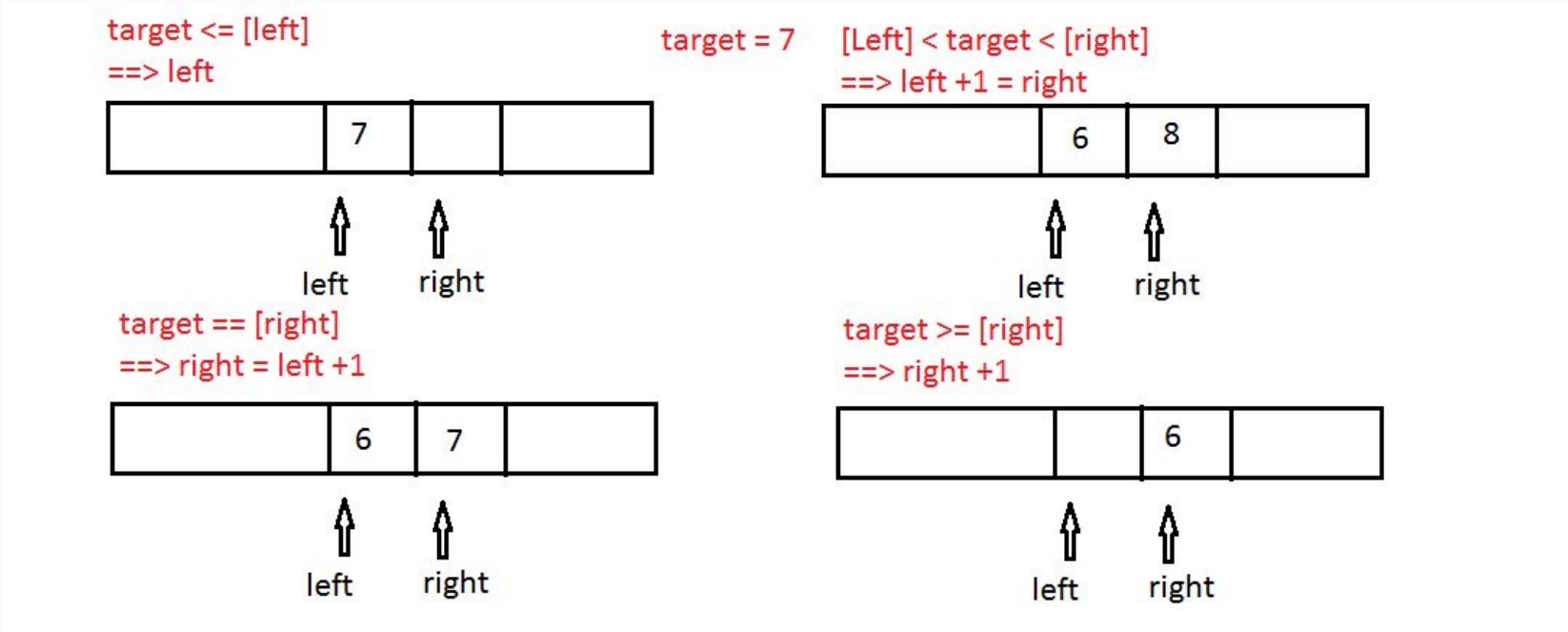
First position:
if ( mid == target) right = mid

Last position:
if ( mid == target) left = mid


代码魔术师
magiciendecode.fr

# 35. Search Insert Position

https://leetcode.com/problems/search-insert-position/



# 69.Sqrt(x)

https://leetcode.com/problems/sqrtx/

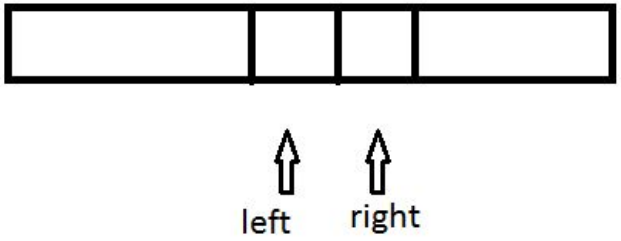
4 = 2 * 2

8 = 2.8... * 2.8... ==> 2

left = 0

// Int.MAX_VALUE = 2147483647 Sqrt(~) = 46340.95...

right = 46340




target = 8

left = 2 , right = 3

target = 8

left = 1 , right = 2

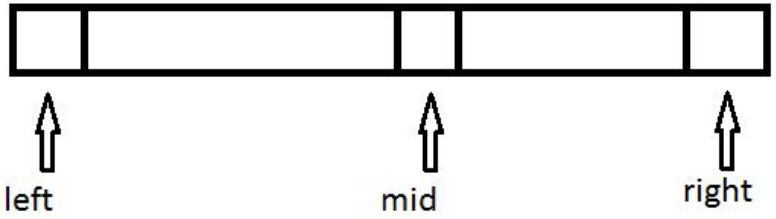
right * right <= target

==> right <= target / right

==> right

# 167. Two Sum II - Input array is sorted

https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/



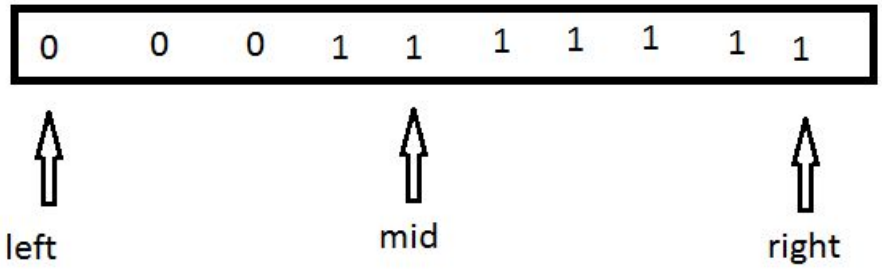

Case 1: [left] + [right] < target ==> left = mid
mid? ==>
[mid] + [right] >= target
~~1 3 5 7 9~~ 11 13 14 15
target = 25 (14 + 11)
[right] = 15 [mid] = 11

Case 2: [left] + [right] > target ==> right = mid
mid? ==>
[mid] + [left] <= target
0 3 5 7 9 11 ~~13 14 15~~
target = 12 (3+9)
[left] = 0 [mid] = 11


代码魔术师
magiciendecode.fr

# 278. First Bad Version

https://leetcode.com/problems/first-bad-version/

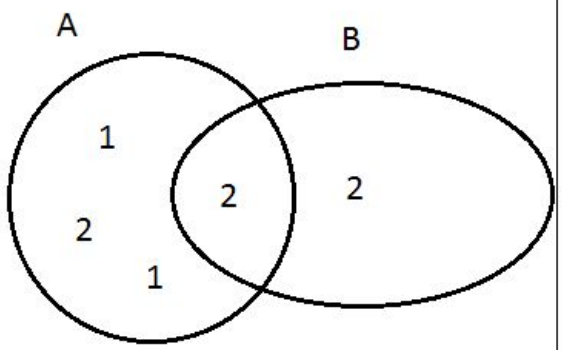



mid isBadVersion ==> right = mid

mid ! isBadVersion ==> left = mid

# 349. Intersection of Two Arrays & 350. Intersection of Two Arrays II

https://github.com/MagicienDeCode/LeetCode_Solution

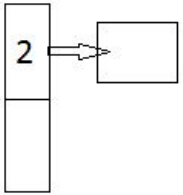


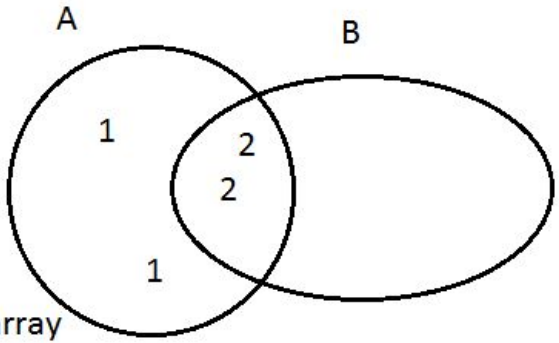

HashSet for small array

HashSet for big array

result array

HashMap < Int, Int>



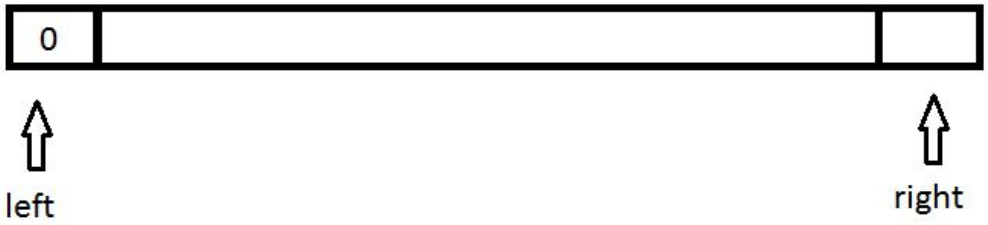

Key = element in small array
value = total number the element in small array

Foreach element in big array, search if element present in map,
if yes and the count > 0, add it to result.
O ( n + m )


代码魔术师
magiciendecode.fr

# 367. Valid Perfect Square

https://leetcode.com/problems/valid-perfect-square/

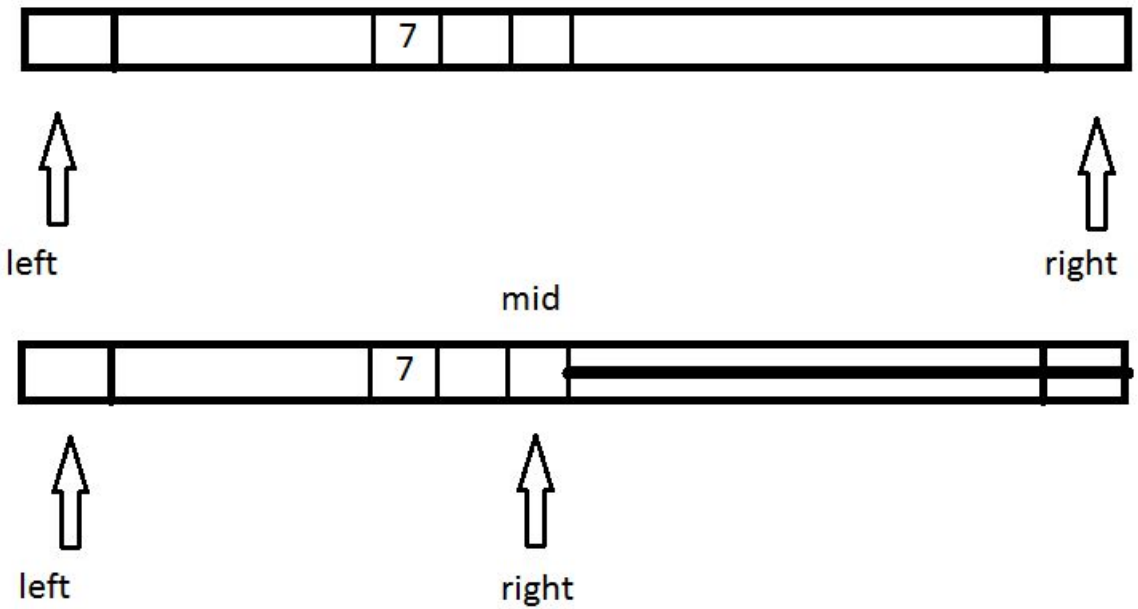



mid * mid == target

right * right == target

# 374. Guess Number Higher or Lower

https://leetcode.com/problems/guess-number-higher-or-lower/





代码魔术师
magiciendecode.fr

# 392. Is Subsequence

https://leetcode.com/problems/is-subsequence/

a h b g d c

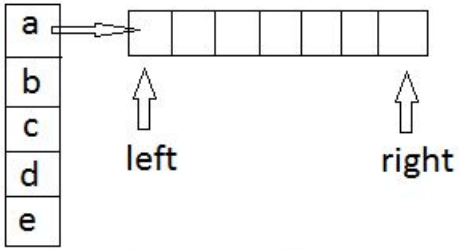


indexOfS = 0
foreach char in t:
if (char == s[indexOfS ]){
indexOfS++
}

indexOfS ==? s.length

O(t)

Follow up O(k*t)



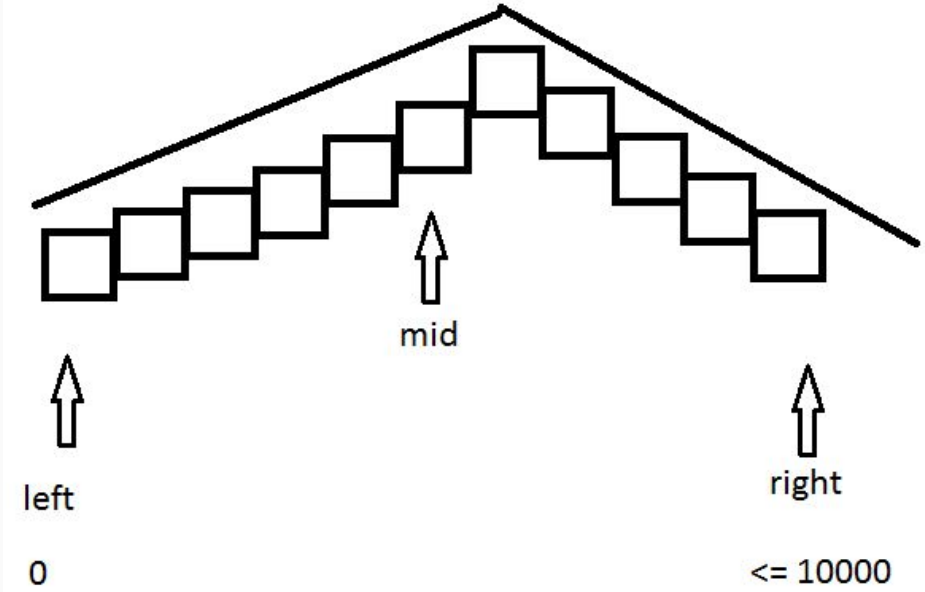

foreach S in S1...Sk:
foreach char in S:
binarysearch in mans.values
a {100,323,4434,32333...}
b {23,56,136,456,7237...}
C {2,65,154,232,1511,6251...}

O(k*100*log(500 000 / 26))


代码魔术师
magiciendecode.fr

# 852. Peak Index in a Mountain Array & 162. Find Peak Element

https://leetcode.com/problems/peak-index-in-a-mountain-array/




[mid-1] < [mid] > [mid+1] ==> return mid
[mid-1] < [mid] < [mid+1] ==> left = mid
[mid-1] > [mid] > [mid+1] ==> right = mid

[left-1] < [left] > [left+1] ==> left
else right

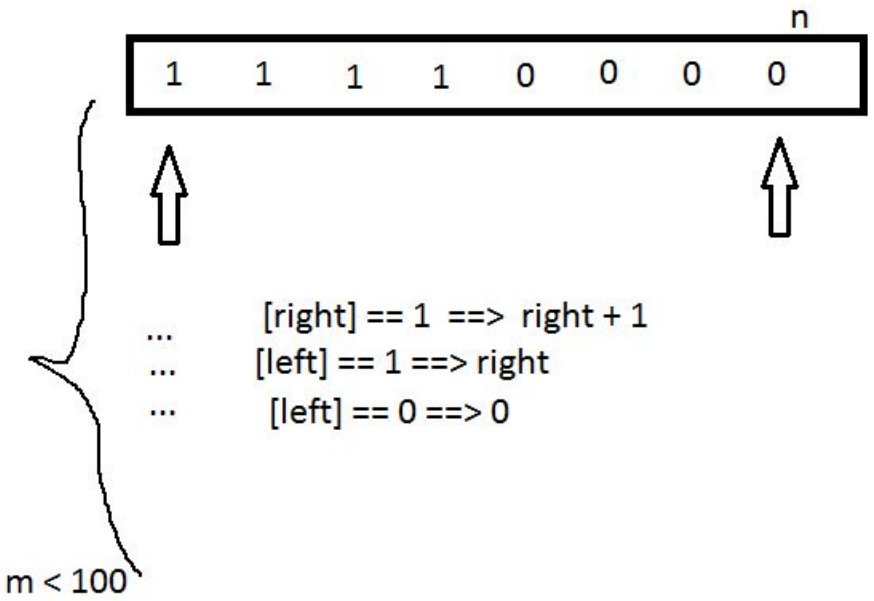
[mid-1] < [mid] ==> left = mid
else -> right = mid

[left] > [right] ==> left
else ==> right


代码魔术师 magiciendecode.fr

# 1337. The K Weakest Rows in a Matrix

https://leetcode.com/problems/the-k-weakest-rows-in-a-matrix/




list < Pair < Int, Int > >

count (1) : total 1 in this line

index: 0 ~ m

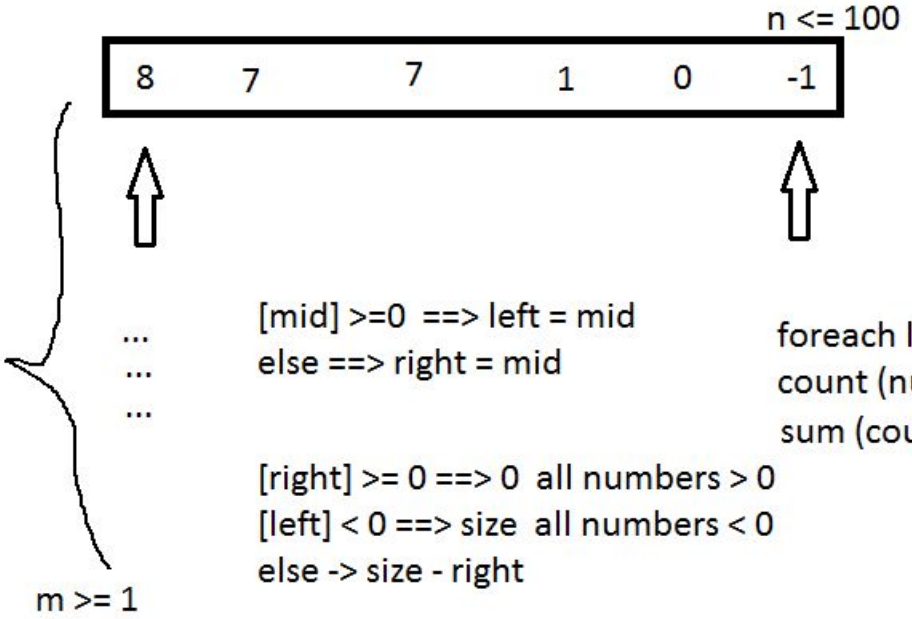
sort by Pair.second

get subList 0,k then return Pair.first


代码魔术师 magiciendecode.fr

# 1351. Count Negative Numbers in a Sorted Matrix

https://leetcode.com/problems/count-negative-numbers-in-a-sorted-matrix/




[mid] >=0 ==> left = mid
else ==> right = mid

[right] >= 0 ==> 0 all numbers > 0
[left] < 0 ==> size all numbers < 0
else -> size - right

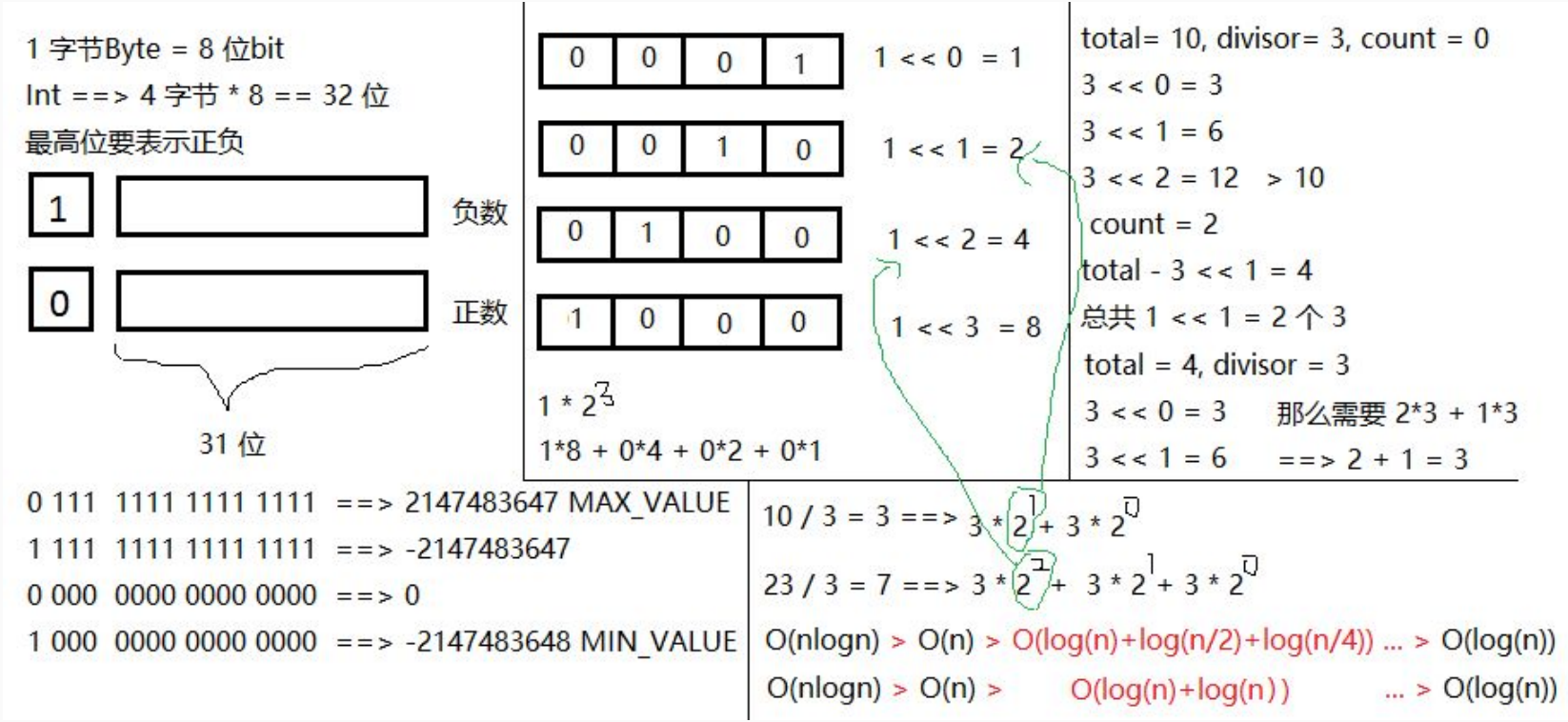
foreach line:
count (number) < 0
sum (count)

fonctional programming:

is a programming paradigm that treats computation as the evaluation of mathematical functions and avoids changing state and mutable data.

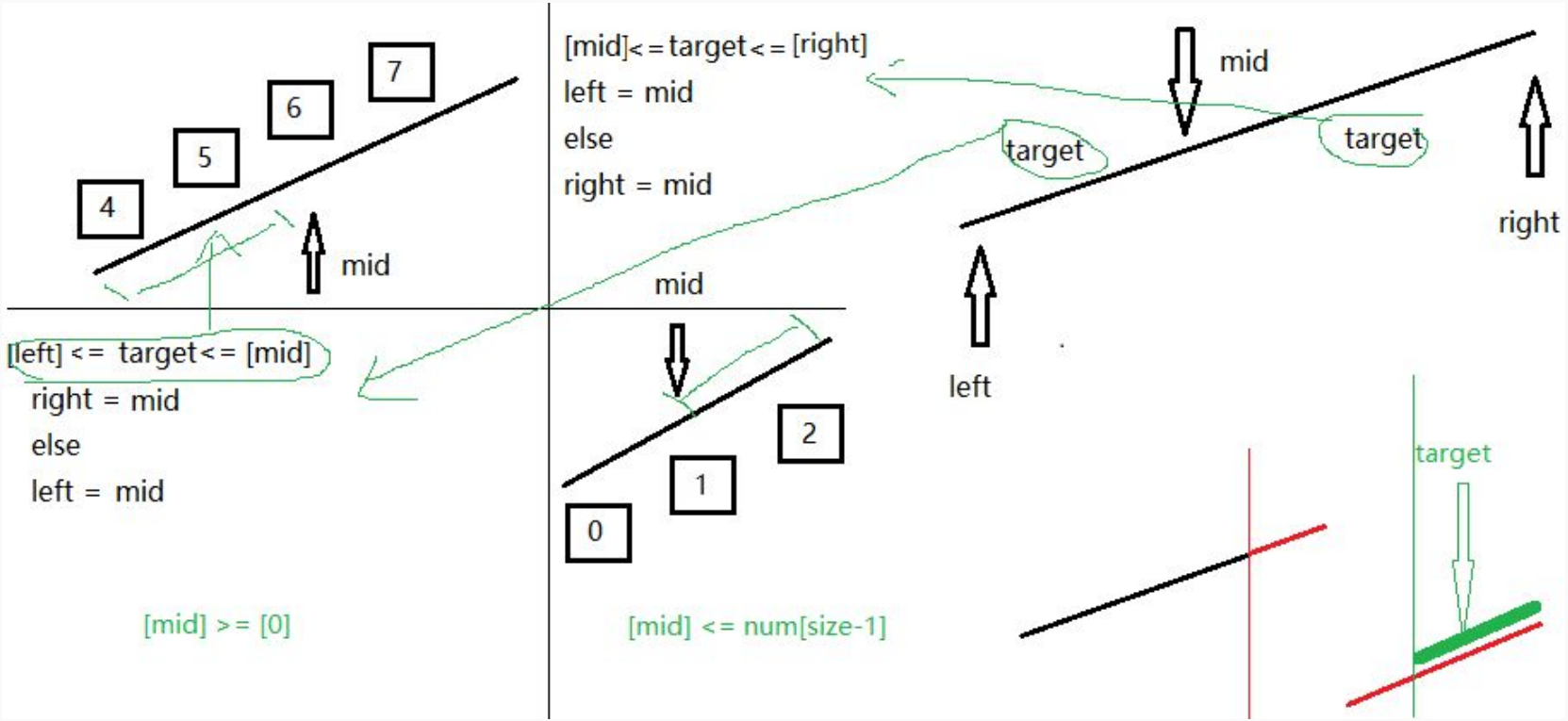
# 29. Divide Two Integers

https://leetcode.com/problems/divide-two-integers/



# 33. Search in Rotated Sorted Array

https://leetcode.com/problems/search-in-rotated-sorted-array/



# 50. Pow(x, n)

https://leetcode.com/problems/powx-n/

$x^n = x * x * x * x * x * x * x * x * x * x * x * x * x * x * x \cdots$ (n)

n % 2 == 0

$x^n = x^{n/2 + n/2} = x^{n/2} * x^{n/2}$

n % 2 == 1

$x^n = x^{(n-1)/2 + (n-1)/2 + 1}$

$= x^{(n-1)/2} * x^{(n-1)/2} * x$

$3^{16} =$ 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3

3 * 3 * 3 * 3 * 3 * 3 * 3 * 3 * 3

3 * 3 * 3 * 3 * 3

3 * 3

3

$3^{16} = 3^8 * 3^8$

$3^8 = 3^4 * 3^4$

$3^4 = 3^2 * 3^2$

O (n) ==> O ( log(n) )

# 74. Search a 2D Matrix

https://leetcode.com/problems/search-a-2d-matrix/

```
matrix = [
        [ 1 ,   3,     5,     7 ],
        [ 10 ,  11,    16,   20 ],
        [ 23 ,  30,    34,   50 ]
]
```

1. find target is in which row:
binary search
O( log(m) )



2. find target in row:
binary search



O( log(n) )

1. verify the given array is not null or empty

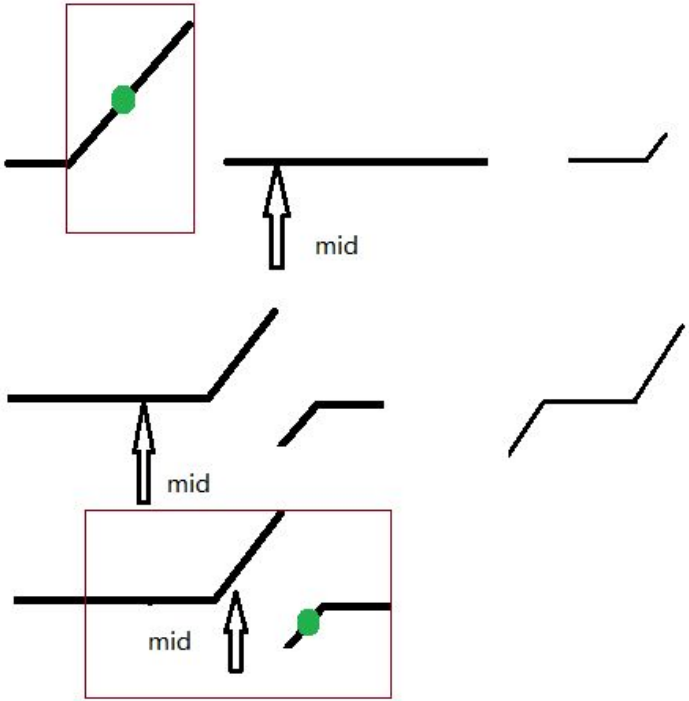
2. relation between mid / target

3. at the end, left + 1 == right,
return left or right or -1


代码魔术师
magiciendecode.fr

# 81. Search in Rotated Sorted Array II

https://leetcode.com/problems/search-in-rotated-sorted-array-ii/




Can't use template "Search in Rotated Sorted Array",
because we do not know should ignore which part if the value of mid is equal to left or right.

So the first step is to pass the duplicate value,
when the value of mid is equal to left, left ++
or it's equal to right, right --

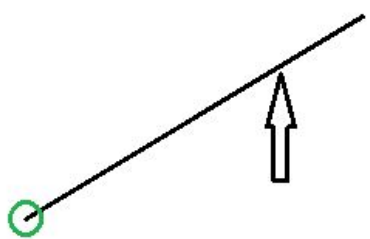
magiciendecode.fr

https://magiciendecode.fr

# 153. Find Minimum in Rotated Sorted Array & 154 II

https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/

https://leetcode.com/problems/find-minimum-in-rotated-sorted-array-ii/

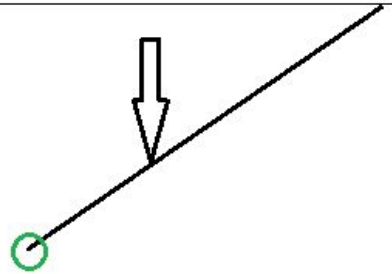


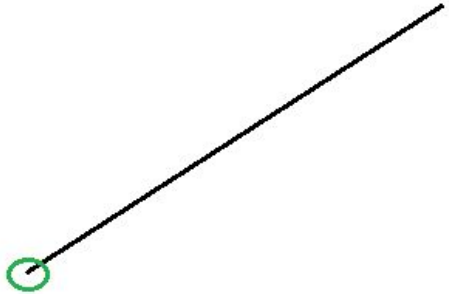
[mid] < [size-1]

right = mid

[mid] > [0]

if [mid] < [size-1]
[0]
else
left = mid

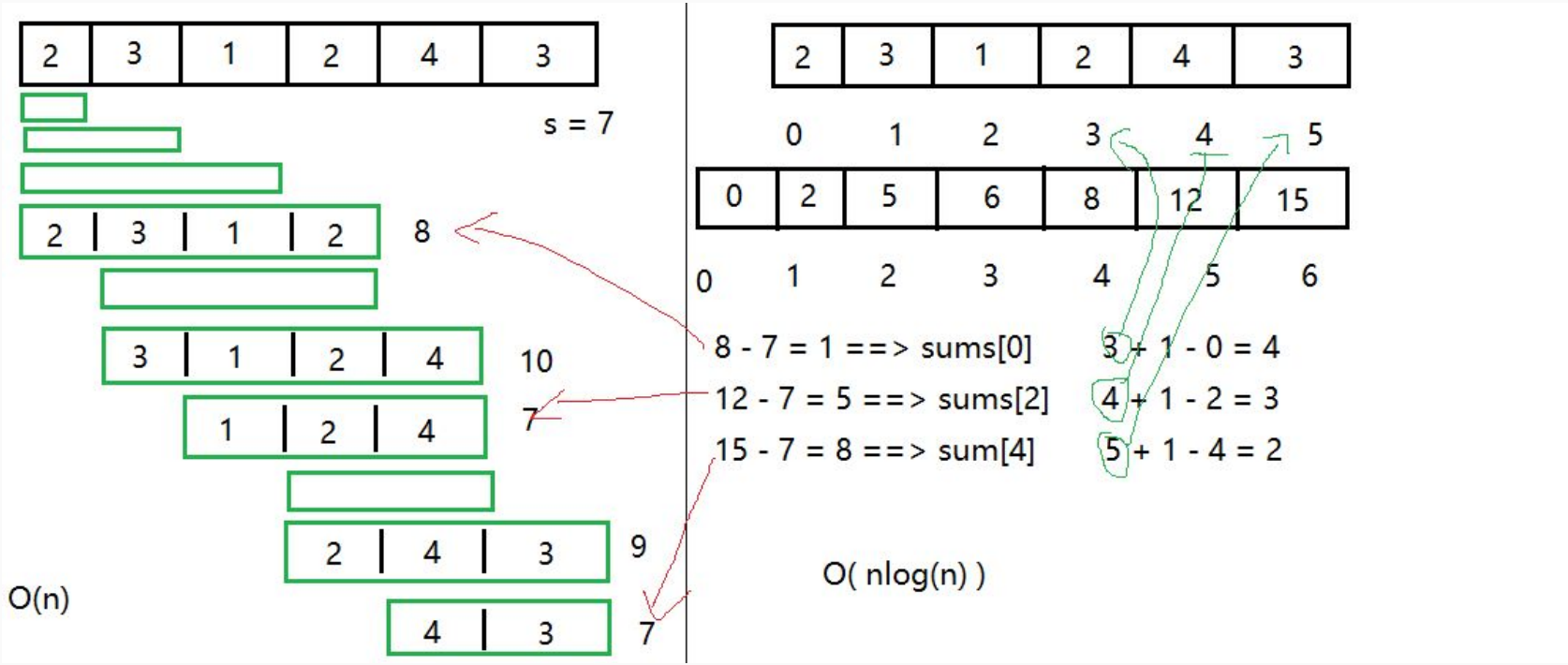




if the array has duplicate items,
first step we should pass them,
then use template.

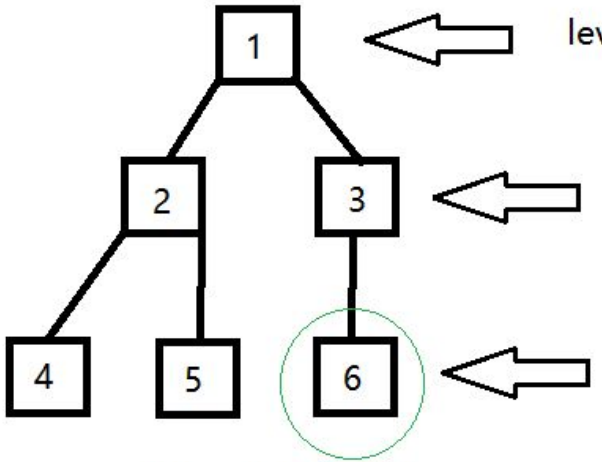
# 209. Minimum Size Subarray Sum

https://leetcode.com/problems/minimum-size-subarray-sum/



# 222. Count Complete Tree Nodes

https://leetcode.com/problems/count-complete-tree-nodes/




level 0 => [1.shl(0),1.shl(1)-1]

level 1 => [1.shl(1),1.shl(2)-1] => [2,3]

level 2 => [1.shl(2),1.shl(3)-1] => [4,7]

left = 1.shl(2) = 4

right = 1.shl(3) -1 = 7

mid = 11 / 2 = 5

target = 6

target > mid ==> RightNode

else ==> LeftNode

Binary search in the last level of tree.

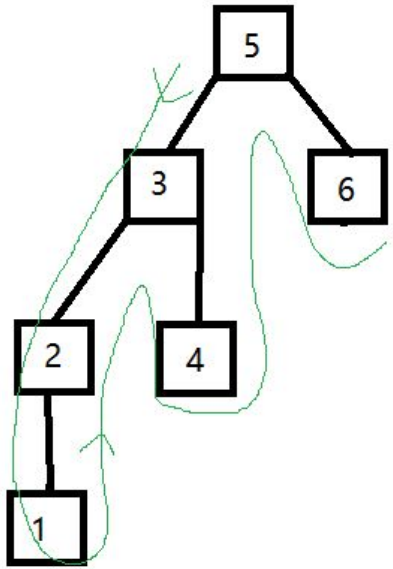
verify the last present node.



代码魔术师
magiciendecode.fr

# 230. Kth Smallest Element in a BST

https://leetcode.com/problems/kth-smallest-element-in-a-bst/




iterate in order: [ 1, 2, 3, 4, 5, 6 ] S(n) O(n)

search the most left node and then add it in result array.

same idea, iterate in order but return the value if it is the kth smallest element. S(1) O(n)

count the sum of left nodes then add itself . O( nlog(n) )

if k == total, return current

if k > total, it means kth element in the right side of current node

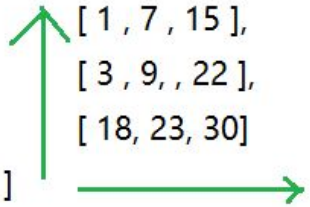
else k < total, result in left side.


代码魔术师 magiciendecode.fr

# 240. Search a 2D Matrix II

https://leetcode.com/problems/search-a-2d-matrix-ii/

matrux = [



[ 1 , 7 , 15 ],

[ 3 , 9, , 22 ],

[ 18, 23, 30]

]

from the bottom left, verify the current and target

if current > target ==> move to previous line

if current < target ==> move to next column

O ( m +n )

Kotlin

fold : Accumulates value starting with [initial] value and applying [operation] from left to right to current accumulator value and each element.

reduce : Accumulates value starting with the first element and applying [operation] from left to right to current acc and earch element.



代码魔术师 magiciendecode.fr

# 274. H-Index && 275. H-Index II

https://leetcode.com/problems/h-index/ https://leetcode.com/problems/h-index-ii/

[ 3, 0, 6, 1, 5 ]

从0到数组的长度(包括)，找到一个值target满足在这个数组中比target大的至少有target个。

for each [ 0 ... size ]: binary search O( log(n) )
verify if it is a valid target.

O( nlog(n) )

validate target: iterate array, count how many element >= target, if total >= target means it's a valid h-index. n

---

[ 0, 1, 3, 5, 6 ]

O( log(n))

validate target use binary search ==> O( log(n) )

as the citations are in ascending order,
citations[length - mid] means at least mid times, then verify it >= mid?
if yes, left = mid and find the largest h-index, if not right = mid


代码魔术师 magiciendecode.fr

# 287. Find the Duplicate Number

https://leetcode.com/problems/find-the-duplicate-number/

[ 1, 3, 4, 2, 2 ]
[ 3, 1, 3, 4, 2 ]

O( n*n ) : foreach element in array: iterate array find if there exists anohter element equals to itself.

O( n* log(n) ): binary search from 0 to size-1,
if the number of element ( <= mid ) <= mid, means no duplicate <= mid, so left = mid
else right = mid. [ 1, 2, 2 ]

O (n): two pointers: fast pointer and slow pointer will meet two times, the second time is the duplicate value. After first meet, fast pointer will go to the begining and move as slow pointer.

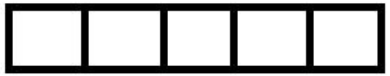

代码魔术师
magiciendecode.fr

# 300. Longest Increasing Subsequence

https://leetcode.com/problems/longest-increasing-subsequence/

[ 10, 9, 2, 5, 3, 7, 101, 18 ]  2 -> 5 ->7 -> 101  2 -> 3 -> 7 -> 101  2 -> 3 -> 7 -> 18

[ 10, 9, 2, 5, 3, 4 ]  ~~2 -> 5~~  2 -> 3 -> 4



this is an array to record longest increasing subsequence.

maxLIS

10

9

2

2 -> 5

2 -> 3

2 -> 3 -> 4

foreach element in original array, try to find the first element >= current, then replace it.

if we use two foreach nested, O( n*n)

if we use foreach and binray search to find the first element in maxLIS

O ( n*log(n) )

# 378. Kth Smallest Element in a Sorted Matrix

https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/

```
matrix = [
    [ 1,  5,  9 ]
    [ 10, 11, 13 ]
    [ 12, 13, 15 ]
]
```

create a descending PriorityQueue.
foreach element in matrix, add it in queue, if size of queue > k, poll an element.
as it's in descending, this queue always keeps k element, so at the end, poll the result that is kth smallest element.
O( n*n*log(k) )

```
[
    [ 1, 2 ]
    [ 1, 3 ]
]
```

Binary search ==> O( log(max-min)*n*log(n) )
left = matrix[0][0]  right = matrix[max][max]
every time count the element that <= mid, if the count < k means kth smallest is between mid and right, so left = mid

how to count?  foreach line, binary search the last element <= target


代码魔术师
magiciendecode.fr

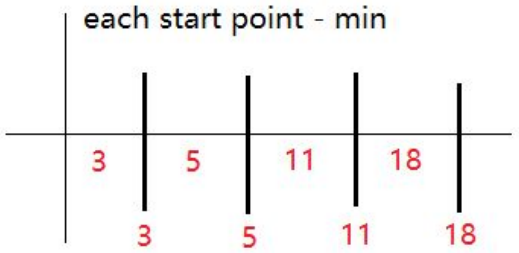
# 436. Find Right Interval

https://leetcode.com/problems/find-right-interval/

[
[ 3, 4 ]
[ 2, 3 ]
[ 1, 2 ]
]

foreach line in matrix: find
foreach line in matrix: compare
O ( n*n )

对于当前矩阵的每一行 I，找到是否存在其他行 J，J的第一个元素[0]
大于等于当前行I的最后一个元素[size-1]，找到差值最小的那一行 J。
对于每一行，找到则返回J在矩阵中的下标，否则-1。
[3,4] doesn't exist one row that [0] >= 4, so return -1
[2,3] ==> index 0 [3,4] 3>=3, so return 0
[1,2] ==> index 1 [2,3] 2>=2, so return 1
The result of example is [-1,0,1]




min of [0]

Then just get [size-1] - min

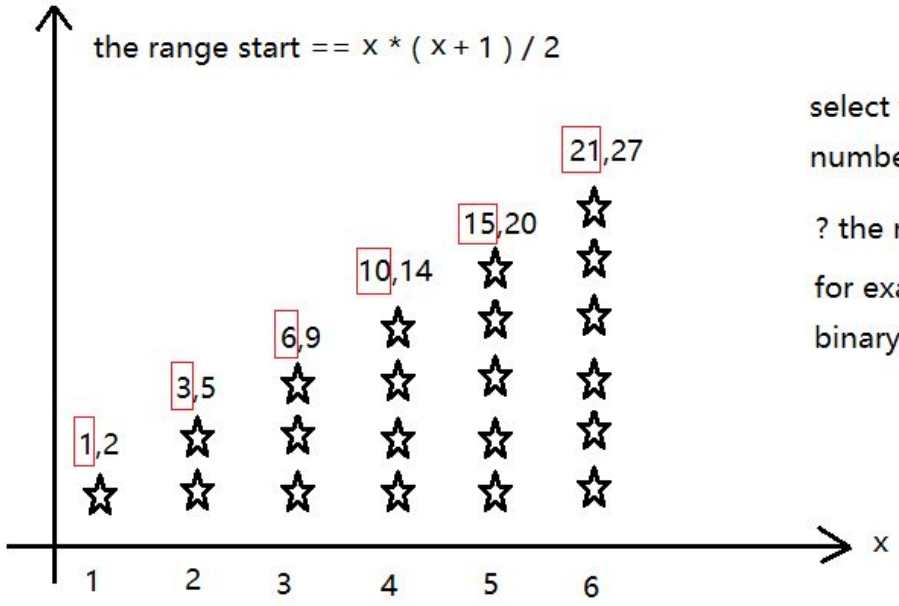
O( 3n + max - min )

S( max-min )


代码魔术师
magiciendecode.fr

# 441. Arranging Coins

https://leetcode.com/problems/arranging-coins/




select from [ 1 to ? ] that x*(x+1)/2 is largest number <= the given n.

? the right side could be n or n / 2

for example, given the n = 10,
binary search range should be 1 to 5.

O ( log(n/2) )


代码魔术师 magiciendecode.fr

# 454. 4Sum II

https://leetcode.com/problems/4sum-ii/

1. use cartesian product of A,B,C,D that will make time complexity n*n*n*n

2. sort D, use cartesian product of A,B,C, search the start and end of target in D. n*log(n) + n*n*n*2log(n)

3. store the cartesian product of C,D , then sort it. n*n + n*log(n)
foreach element in cartesian product of A,B, find the start and end of target in sortedCD. n*n*2log(n*n)

4. use hashMap but this will take more memory. n*n + n*n

cartesian product of A,B,C,D
each array has two elements

A[0] + B[0] + C[0] + D[0]
A[0] + B[0] + C[0] + D[1]
A[0] + B[0] + C[1] + D[0]
A[0] + B[0] + C[1] + D[1]
A[0] + B[1] + C[0] + D[0]
A[0] + B[1] + C[0] + D[1]
A[0] + B[1] + C[1] + D[0]
A[0] + B[1] + C[1] + D[1]
A[1] + ...

# 475. Heaters

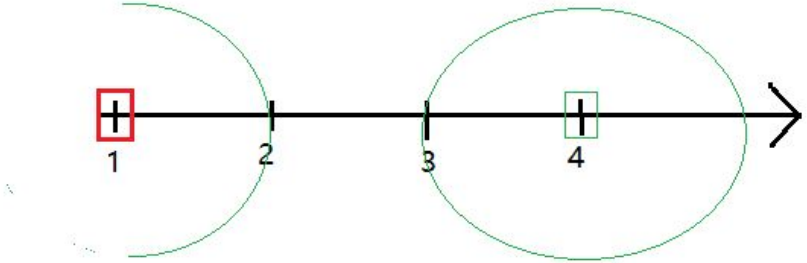
https://leetcode.com/problems/heaters/

array A [ 1, 2, 3, 4 ]

array B [ 1, 4 ]

foreach element in array A find an element in B that make the minimum difference value between them.

Then for all these minimum differences, find the maximum.




binary search in array B

if B empty, return 0
if house < [0] , return [0] - house
if house > [max], return house - [max]

[mid] = house means distance 0

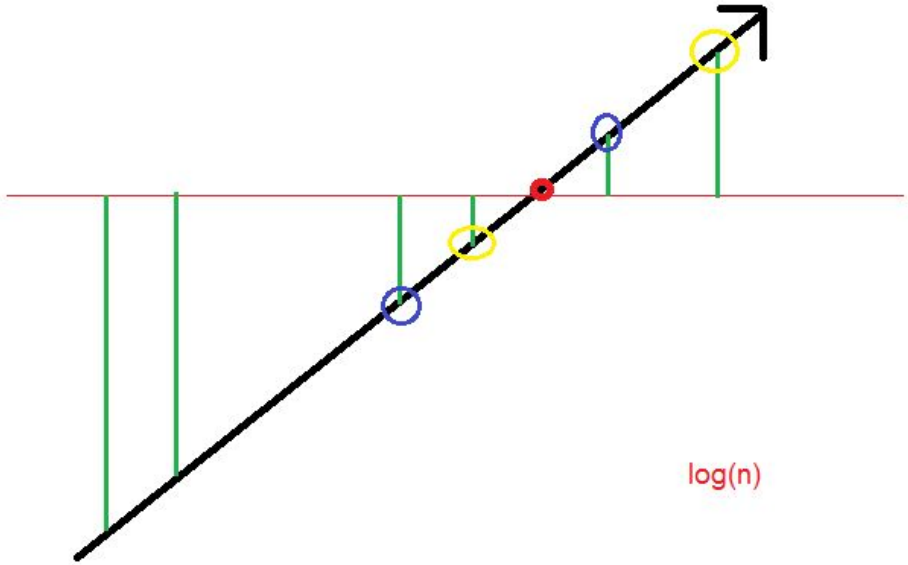
at the end, we will find
[left] < house < [right]

return minOf house - [left] and [right] - house


代码魔术师
magiciendecode.fr

# 658. Find K Closest Elements

https://leetcode.com/problems/find-k-closest-elements/




binary search in array an element that most close to target, then base on the index, take index - 1 or index + 1 according to the distance between the value and target.

log(n)*k

binary search in array the start index

if target - [mid] > [mid+k-1] - target, left = mid
else right = mid

we use our template, at the end we find left,right
take an example, the blue point and yellow point.

just compare the distance between target - [left] and [right+k-1] - target


代码魔术师 magiciendecode.fr

# 718. Maximum Length of Repeated Subarray

https://leetcode.com/problems/maximum-length-of-repeated-subarray/

A [ 1, 2, 3, 2, 1 ]
B [ 4, 3, 2, 1, 7 ]

```
foreach in array A:
    foreach in array B:
        if(a==b)
            while
            count if next elements equals
```

n * n * n

foreach in A:
1, 2, 3, 2, 1
4, 3 ,2 ,1, 7 ==> 0
2, 3, 2, 1
4, 3, 2, 1, 7 ==> 3
3, 2, 1
4, 3, 2, 1, 7 ==> 0

2 * n * n

3,2,1
4,3,2,1,7

foreach in B do the same things with A

dynamc programming, using a memory-based data structure to avoid repeated computing.

S ( n * n )
O ( n * n )

| | 1 | 2 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| 4 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 0 | 1 | 0 | 0 |
| 2 | 0 | 1 | 0 | 2 | 0 |
| 1 | 1 | 0 | 0 | 0 | 3 |
| 7 | 0 | 0 | 0 | 0 | 0 |

binary search in [1,1000]
but how to verify mid is a valid length?

convert A and B to string,
first you should map it to char, why?
we want to use substring and hashSet to verify if exists a length of string that make a subarray.
think about 12,4,3, if we use int, 1243

O ( (2n) * log 1000 )

then we should create a hashSet, then put S(n)
substrng(index, index + targetLength) of A in this hashSet, O(n)
foreach substring(index, index + targetLength) of B, verify if hashSet contains it. O(n)

# 875. Koko Eating Bananas

https://leetcode.com/problems/koko-eating-bananas/

1 ... ... ... 1000000000

O( n * log(1000 000 000) )

binary search to find the first valid target
XXXXXXXXXXOOOOOOOO

A valid target means
foreach element in array
calculate i / target + (1 or 0, according to i%target)
then sum the result, compare with H, if result <= H, it's a valid target.

int left = (int)(sum / H)
left = left == 0 ? 1 : left
int right = (int)(sum / (H - piles.length +1)) + 1

missing testcase:
1 <= piles.length <= 10 000
piles.length <= H <= 1 000 000 000
1<= piles[i] <= 1 000 000 000

wrong code:

int left = (int)(sum / H) + 1

Input:
piles = [2,2]
H = 2

expected k = 2

but answer is 3


代码魔术师 magiciendecode.fr

# 528. Random Pick with Weight

https://leetcode.com/problems/random-pick-with-weight/

weight [ 1, 3, 6]
it means if you select from index 0,1,2
that you have 10% fall on index 0,
30% fall on index 1
60% fall on index 2

This question can be transformed as

given an array and a target in [1,array.max]
return the first index that array[index] >= target

1. calculate sum [1,4,10]

XXXXXXOOOOOOO

1 2 3 4 5 6 7 8 9 10

in kotlin

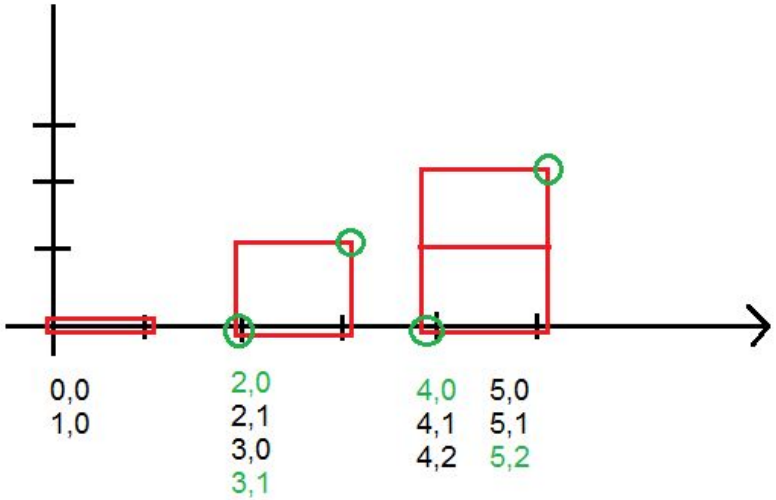
init{...} will be executed after the main constructor

if there are several init bloc, they will be executed in order.


代码魔术师
magiciendecode.fr

# 497. Random Point in Non-overlapping Rectangles

https://leetcode.com/problems/random-point-in-non-overlapping-rectangles/




[2,4,6]

weight [ 1, 3, 6]
it means if you select from index 0,1,2
that you have 10% fall on index 0,
30% fall on index 1
60% fall on index 2

given an array and a target in [1,array.max]
return the first index that array[index] >= target

XXXXXXOOOOOOO

if we find know the point should at which rectangle,
then randomly return a [x,y] in it.
for example the rectangle[4,0,5,2]
the x of point could be 4 + random.nexInt(5-4+1)
the y of point could be 0 + random.nextInt(2-0+1)

# 911. Online Election

https://leetcode.com/problems/online-election/

```
index      0    1    2    3
persons [  3    0    3    1    ]
times   [  5    12   39   100  ]
```

```
leaders[ persons[0], persons[1], persons[0], persons[0] ]
           times[0]   times[1]   times[2]   times[3]
```

The question is given a time (for example 7 )
in [ times[0], times[max] ] ==> [5,100]
return the person that at time 7 get the most vote.

counts int[] max length is persons.length as 0 <= persons[i] <= persons.length
// at index i, the value means person[i]'s count

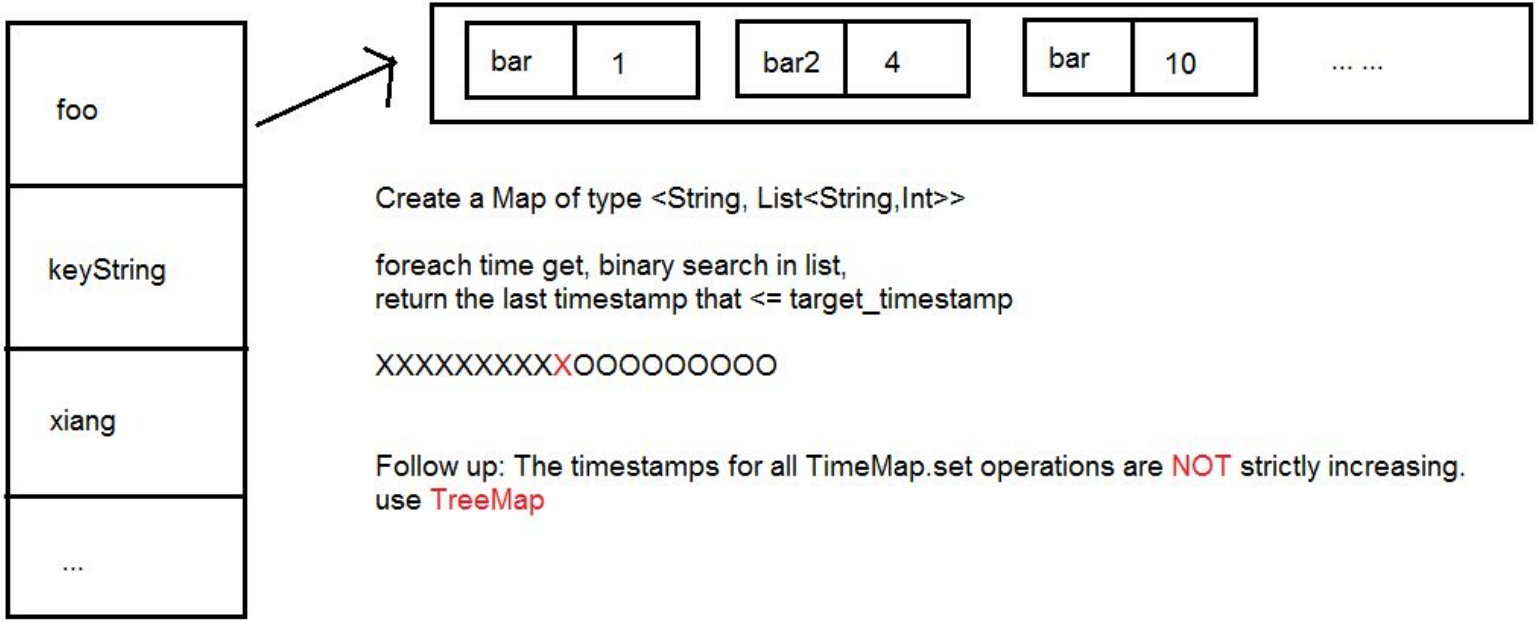
leaders int[] max length is times.length S(n)
// at index i, time[i]'s leader

given a random time, use binary search return the last index that time[index] <= target 10 000 * log(n)
then return leaders[index] which represent the leader at time[index]


代码魔术师
magiciendecode.fr

# 981. Time Based Key-Value Store

https://leetcode.com/problems/time-based-key-value-store/




Create a Map of type <String, List<String,Int>>

foreach time get, binary search in list,
return the last timestamp that <= target_timestamp

XXXXXXXXXOOOOOOOOO

Follow up: The timestamps for all TimeMap.set operations are NOT strictly increasing.
use TreeMap

# 1011. Capacity To Ship Packages Within D Days

https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/

binary search, first we should set left and right,
let's search the max weight in array, then set left = max_weight
the right won't be exceed max_weight * max_weights_length, so right = max_weight * 500 50000

Then this quesiton transform to select the first valid capacity from [ max_weight, 500 * max_weight ]

XXXXXXXXXLLLLLLLLLLL

days = 1, count = target

```
foreach w in weights
        count = count - w
        if( count < 0 ){
                days++
                count = target - weight
        }
```

weights [ 3, 2, 2, 4, 1, 4 ]  D = 3, target = 5

day 1 [3,2] count = 0
day 2 [2]  0-2 < 0 ==> count = 5 - 2 = 3
day 3 [4]  3 - 4 < 0 ==> count = 5 - 4 = 1
[1]  1 - 1 = 0
day 4 [4]

# 1300. Sum of Mutated Array Closest to Target

https://leetcode.com/problems/sum-of-mutated-array-closest-to-target/

```
sum = 0

foreach element in array:
        sum += minOf(element,?)
```

[ 4, 9, 3 ] , target = 10
[ 3, 3, 3 ] ==> 9

[ 2, 3, 5 ] , target = 10
[ 2, 3, 5 ] ==> 10

[ 4, 9, 3 ] , target = 30
[ 4, 9, 3 ] ==> 16

[ 4, 9, 3 ] , target = 6
[ 2, 2, 2 ] ==> 6
The maximum available value is target

[ 4, 9, 3 ] , target = 0
[ 0, 0, 0 ] ==> 0
The minumum available value is 0

```
Binary search from 0 to target
        if sum == target, return it
        if sum < target
                left = mid
        if sum > target
                right = mid
```

O( n * log(target) )

# 1201. Ugly Number III

https://leetcode.com/problems/ugly-number-iii/

left = minOfabc
right = minOfabc * n

binary search : calculate the mid is ?th ugly number

mid / a + mid / b + mid / c
- mid / lcmab - mid / lcmac - mid / lcmbc
+ mid / lcmabc

n = 1000000000
a = 2
b = 217983653
c = 336916467

1999999983 ==> 999999999
1999999984 ==> 1000000000
1999999985 ==> 1000000000

LCM, Least Common Multiple 最小公倍数
a * b / gcd(a,b)

GCD Greatest Common Divisor 最大公约数
辗转相除法
given a,b each time ensure a > b, then if a%b == 0 return b
else gcd(b,remainder)

319,377
1 ==> 377 % 319 ... 58
2 ==> 319 % 58 ... 29
3 ==> 58 % 29 ... 0
return 29

更相减损法

63,98
98 - 63 = 35
63 - 35 = 28
35 - 28 = 7
28 - 7 = 21
21 - 7 = 14
14 - 7 = 7
return 7

# 1283. Find the Smallest Divisor Given a Threshold

https://leetcode.com/problems/find-the-smallest-divisor-given-a-threshold/

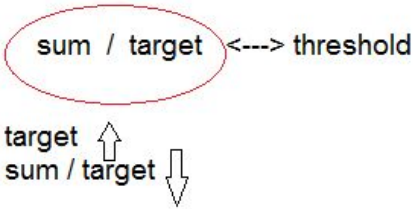
[10] , threshold = 11
left = 1
[ 2, 20 ] , threshold = 1
right = maxOfArray




sum = 0
foreach element in array:
    sum += element / target + ( 1 or 0 )

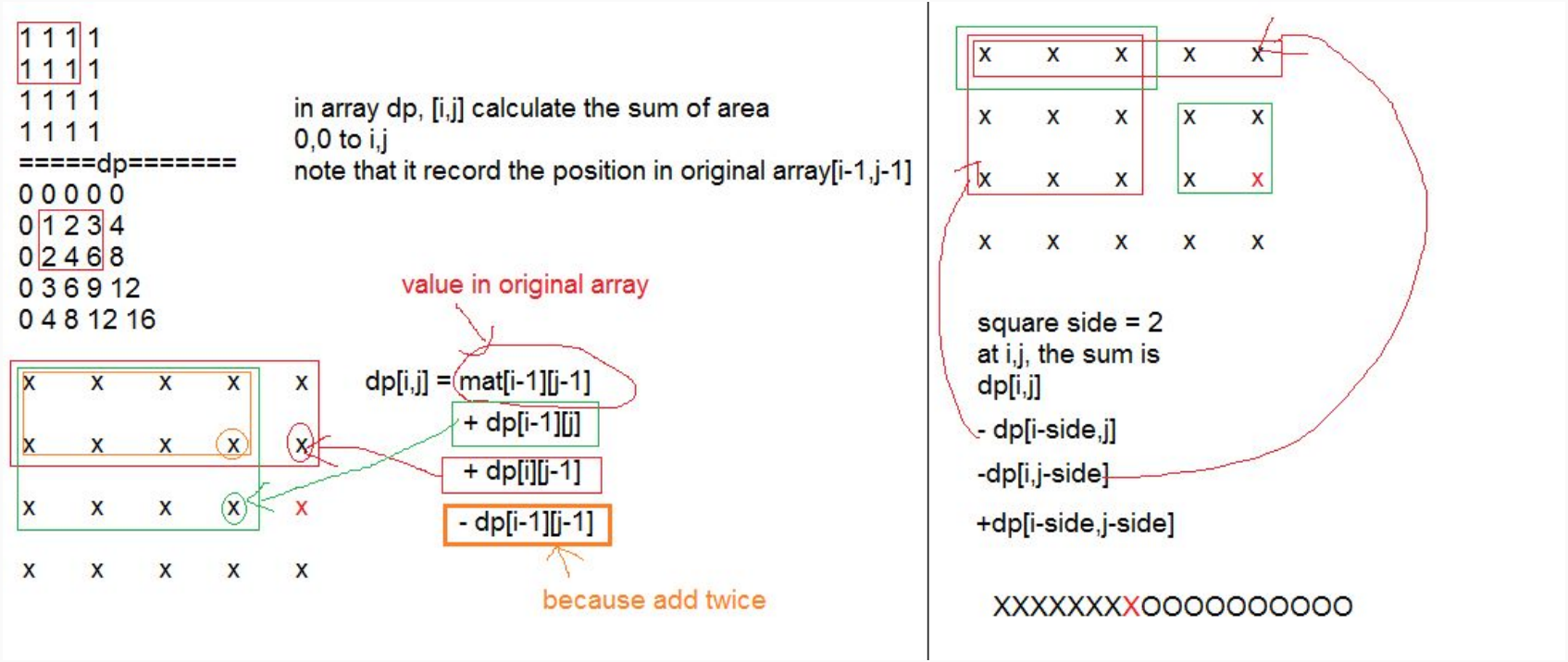
sum == threshold -> return mid ?

sum > threshold -> left = mid
else -> right = mid



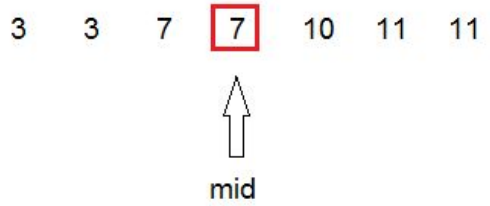
# 1292. Maximum Side Length of a Square with Sum Less than or Equal to Threshold

https://leetcode.com/problems/maximum-side-length-of-a-square-with-sum-less-than-or-equal-to-threshold/



# 540. Single Element in a Sorted Array

https://leetcode.com/problems/single-element-in-a-sorted-array/




1. single element
   previous != single element != next

2. single element in which part

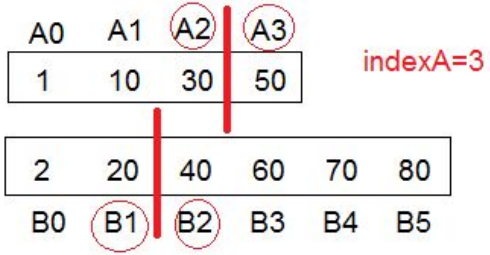
not single element

if single element is in the right part, left = mid
   1. second not single element && index % 2 != 0
   2. first not single element && index % 2 == 0

代码魔术师 magiciendecode.fr

# 4. Median of Two Sorted Arrays

https://leetcode.com/problems/median-of-two-sorted-arrays/

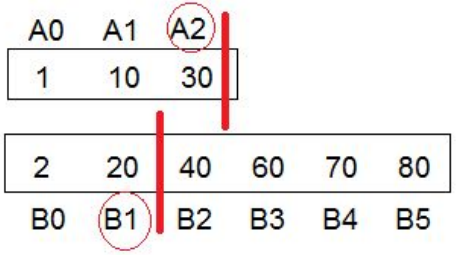



indexA=3

indexB=2

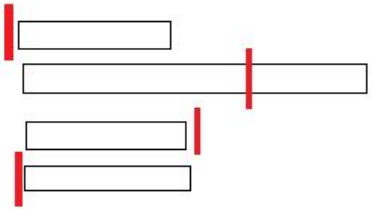
$$\frac{\text{maxOf( A2, B1 ) + minOf( A3, B2 )}}{2.0}$$

$$\frac{\text{maxOf(left_part_AB) + minOf(right_part_AB)}}{2.0}$$




maxOf( A2 , B1 )

maxOf(left_part_AB)



left_part_AB = right_part_AB

max(left_part_AB) <= min(right_part_AB)

indexB = (A+B+1) / 2 - indexA

binary search indexA
indexB
A[indexA-1] < A[indexA]
B[indexB-1] < B[indexB]

if midA < the correct indexA, left = mid
indexB >=1 && B[indexB-1] > A[midA]

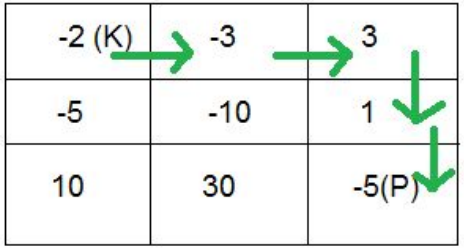
if midA > the correct indexA, right = mid
midA >= 1 && A[indexA-1] > B[indexB]


代码魔术师
magiciendecode.fr

# 174. Dungeon Game

https://leetcode.com/problems/dungeon-game/

dynamic programming




| | | | |
|---|---|---|---|
| 7 (K) | 5 | 2 | max |
| 6 | 11 | 5 | max |
| 1 | 1 | 6 (P) | 1 |
| max | max | 1 | |

S(m)

O(m*n)

how to binary search ?

start from (P), we need K is alive HP = 1
so K should have minimum 1 - (-5) = 6
before entre this room(3,3)

Then calculate each room: the value before enter is dp[i,j] = 1 or ( minOf(dp[i+1,j], dp[i,j+1]) - dungeon[i][j] )

DOWN RIGHT

the mimimum HP for next step

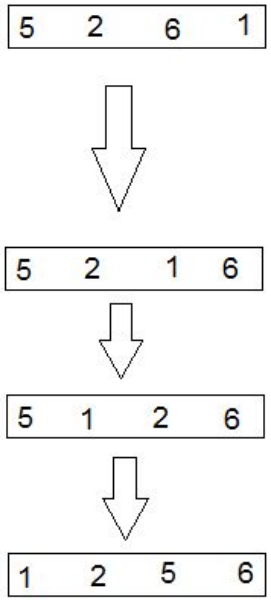
-

current value

if the result is negative means current room can add HP
K just need to be alive HP = 1

# 315. Count of Smaller Numbers After Self

https://leetcode.com/problems/count-of-smaller-numbers-after-self/




| | |
|---|---|
| 1 | nums[3] = 1, in new sorted array, index 0 |
| 1 6 | nums[2] = 6, in new sorted array, index 1 |
| 1 2 6 | nums[1] = 2, in new sorted array, index 1 |
| 1 2 5 6 | nums[0] = 5, in new sorted array, index 2 |

S ( n )
O ( n * log(n) )

binary search the insert position

[mid] <= target ------ left = mid
[mid] > target ------ right = mid

target <= [left] ------ return left
target <= [right] ------ return right
else -> right + 1

[mid] < target ------ left = mid
[mid] >= target ------ right = mid

2 3 3 3 3 9

# 327. Count of Range Sum

https://leetcode.com/problems/count-of-range-sum/

| -2 | 5 | -1 |
|---|---|---|

nums

| 0 | -2 | 3 | 2 |
|---|---|---|---|

sums

sums[i+1] = sums[i] + nums[i]

0 1 2 3

sum of array from [i to j] = sums[j+1] - sum[i]
[0 to 0] = sum[1] - sum[0] = -2
[1 to 2] = sum[3] - sum[1] = 2 - (-2) = 4 <---> 5 + (-1)

```
for i in 0 .. nums.size
        for j in i+1 .. sums.size
                sums[j] - sum[i] ?
```

S (n)
O ( n*n )

iterate all ranges
without loop calculate sum

!! LongArray !!

lower <= sums[j] - sums[i] <= upper sums[j] fix, search range sums[i]
when loop to calculate sums[j]
if sums could be sorted, then find min and max index for sums[i]
count = max - min
sums[i] >= sums[j] - upper <---> binary search first index
sums[i] <= sums[j] - lower <---> binary search last index
in order to keep sorted sums, binary search insert postion
S (n)
O (n * log(n))

lower <= sums[j] - sums[i] <= upper sums[i] fix, search range sums[j]

given the sums[i], let's see sums[j]
sums[j] <= sums[i] + upper
sums[j] >= sums[i] + lower

Merge Sort, when merge, find the range of j

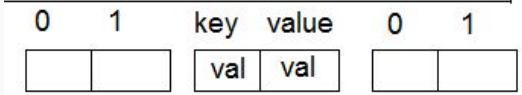
S (n)
O (n * log(n))

# 352. Data Stream as Disjoint Intervals

https://leetcode.com/problems/data-stream-as-disjoint-intervals/

1,3,7,2,6

[1,1]
[1,1],[3,3]
[1,1],[3,3],[7,7]
[1,1],[2,2],[3,3],[7,7] ==> [1,3],[7,7]
[1,3],[6,6],[7,7] ==> [1,3],[6,7]




leftInterval
rightInterval

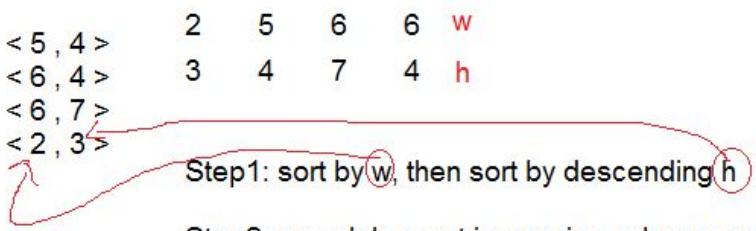
leftInterval
[1] >= val -> return
[1] +1 == val -> key = [0], remove leftInterval
rightInterval
[0] -1 == val, value = [1], remove rightInterval

treeMap.put(key,value)

```
size = 0 -> add(new)                    MutableList<IntArray>
size = 1 -> {
    val +1 < first[0] -> add(0,new) / val + 1 == first[0] -> first[0] = val
    first[1] + 1 == val -> first[1] = val / first[1] + 1 < val -> add(1,new)
}
size > 1 -> {
    targetPosition == binarySearchLastPositionLessThanTarget
    targetPostion == 0 && val < [0] ->{
        val + 1 < [0] -> add(0,new) / val + 1 == [0] -> [0] = val
    }
    targetPosition == size -1 ->{
        [1] + 1 = val -> [1] = val / [1] + 1 < val -> add(size,new)
    }
    1<=targetPosition<=size-2 ->{
    val -1 > left[1] && val +1 < right[0] -> add(targetP+1,new)
    val -1 == left[1] && val +1 < right[0] -> left[1] = val
    val -1 > left[1] && val +1 == right[0] -> right[0] = val
    val -1 == left[1] && val +1 == right[0] ->
        left[1] = right[1], removeAt(targetP+1)
    }
}
```

# 354. Russian Doll Envelopes

https://leetcode.com/problems/russian-doll-envelopes/



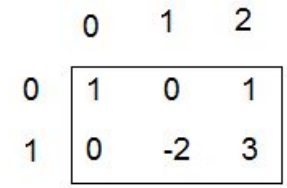

Step1: sort by w, then sort by descending h

Step2: search longest increasing subsequence
https://leetcode.com/problems/longest-increasing-subsequence/

Java :
https://github.com/MagicienDeCode/LeetCode_Solution/blob/master/src/main/java/binary_search/LongestIncreasingSubsequence300.java
Kotlin :
https://github.com/MagicienDeCode/LeetCode_Solution/blob/master/src/main/kotlin/binary_search/LongestIncreasingSubsequenceKotlin300.kt

# 363. Max Sum of Rectangle No Larger Than K

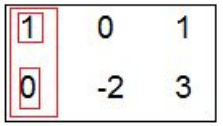
https://leetcode.com/problems/max-sum-of-rectangle-no-larger-than-k/



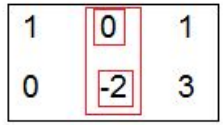

```
for ( col in 0 1 2 )
        rowSum[0]
        rowSum[1]
for (  c   in col to 0 )
        sortedSum.add 0
        currentSum = 0
for ( r in 0 1)
        rowSum[r] += matrix[r][ c ]
        currentSum += rowSum[r]
        verify currentSum - k in sortedSum
             result = maxOf(currentSum - ?)
        sortedSum.add ( currentSum )
```

binary search the first element that greater equal than target

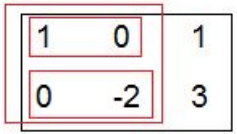
col =0 c =0 r =0 [0,0]
[0,0]
r =1 [1,0]
[0,0][1,0] [1,0][0,0]




col =1 c =1 r =0 [0,1]
[0,1]
r =1 [1,1]
[0,1][1,1] [0,1][1,1]




c =0 r =0 [0,1][0,0]
[0,1][0,0]
r =1 [1,1][1,0]
[0,1][0,0] [1,1][1,0] [0,1][0,0] [1,1][1,0]




代码魔术师
magiciendecode.fr

# 410. Split Array Largest Sum

https://leetcode.com/problems/split-array-largest-sum/

[ 7,2,5,10,8 ]

when m = 1, sum = array.sum()
when m = array.size, sum = max

OOOOOOOOOXXXXXXXXXX

count = 1
a valid sum is

foreach i in array
    sum += i
    if(sum>target)
        sum = i
    count ++
    verify count > aim?

| | 7 | 2 | 5 | 10 | 8 |
|---|---|---|---|---|---|
| 1 | 7 | 9 | 14 | 24 | 32 |
| 2 | 7 | 7 | 7 | 14 | 18 |

when split,
the max of sum is
dp[i-1][?] or sums[j] - [?]

then dp[i][j] = minOf

i = 2

7 2 5

7 | 7
9 | 5

minof = 7

7 2 5 10

7 | 17
9 | 15
14 | 10

minof = 14

7 2 5 10 8

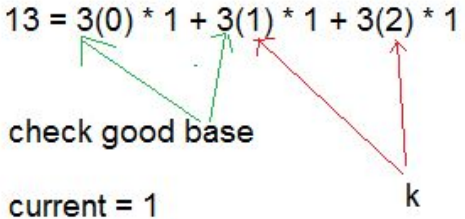
7 | 25
9 | 23
14 | 18
24 | 8

minof = 18

代码魔术师
magiciendecode.fr

# 483. Smallest Good Base

https://leetcode.com/problems/smallest-good-base/




check good base

current = 1
current = 1*3 + 1
current = 4*3 + 1

foreach base from 2 to n
if(good base)
return

nlog(base)(n) ==> Time Limit Exceeded

base = n-1
minK =2

K larger
base smaller

base = 2
maxK = log2(n) +1

from maxK downTo minK
binarySearch valid base

baseLeft = 2
baseRight = ? n

log2(n) * log(base)(n) * log(base)(n)

given K and base and n

if base(0) + ... base(K) == n OK
< n -> left = mid
> n -> right = mid

attention Long overflow

代码魔术师
magiciendecode.fr

# 493. Reverse Pairs

https://leetcode.com/problems/reverse-pairs/

[ 1, 3, 2, 3, 1 ]

i = 1 [i] = 3
j = 4 [j] = 1

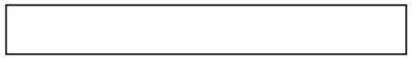
i = 3 [i] = 3
j = 4 [j] = 1

```
for i in 0 .. size -2
        for j in i+1 .. size -1
                verify!!
```

Time Limit Exceeded

n * n

sortedList of j



```
for i in size-1 .. 0
        binarySearch
        then add
```

target > [j].toLong * 2

right ?
left ?
-1

2 * n * log(n)

Merge Sort

it will garanteed the i < j, then just binary search [i] > 2*[j]

n * log(n)

代码魔术师
magiciendecode.fr

# 668. Kth Smallest Number in Multiplication Table

https://leetcode.com/problems/kth-smallest-number-in-multiplication-table/

| | | |
|---|---|---|
| 1 | 2 | 3 |
| 2 | 4 | 6 |
| 3 | 6 | 9 |

binary search from [ 1 to m*n ]

count of element less equal than target

< k ==> left = mid

>= k ==> right = mid

forearch line

binary search lessEqual than target postion

log(m*n) * m * log(n)

forearch line

count +=

max <= target ? n

else target / line

log(n*m) * m

https://github.com/MagicienDeCode/LeetCode_Solution

Ctrl + F ==> 378. Kth Smallest Element in a Sorted Matrix

https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/

代码魔术师
magiciendecode.fr

# 719. Find K-th Smallest Pair Distance

https://leetcode.com/problems/find-k-th-smallest-pair-distance/

k

priority queue with desceding order

```
for i in 0..size-2
        for j in i+1 .. size -1
                distance = abs(nums[i]-nums[j])
```

Memory Limit Exceeded

1. sort

2. binary search from 0 to max-min

3. how to calculate how many distances <= target

1 2 4 10 target = 4

1 2 [0,1]
1 4
2 4 [0,2]

代码魔术师
magiciendecode.fr

# 778. Swim in Rising Water

https://leetcode.com/problems/swim-in-rising-water/

| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 24 | 23 | 22 | 21 | 5 |
| 12 | 13 | 14 | 15 | 16 |
| 11 | 17 | 18 | 19 | 20 |
| 10 | 9 | 8 | 7 | 6 |

BFS + PriorityQueue

1 (24)
2 (23 24)
3 (22 23 24)
4 (21 22 23 24)
5 (21 22 23 24)
16 (21 22 23 24)
...

binary search from min to max in matrix
for{for{}}

We need a matrix visited and we should clean this matrix each time before we calculate the mid.

how to verify if a given target is valid or not? DFS

```
if (inGrid(x, y, grid.size, grid[0].size)
    && visited[x][y] == 0 &&
    grid[x][y] <= target
) {
    visited[x][y] = 1
    dfs(grid, visited, x - 1, y, target)
    dfs(grid, visited, x, y - 1, target)
    dfs(grid, visited, x + 1, y, target)
    dfs(grid, visited, x, y + 1, target)
}
```

代码魔术师
magiciendecode.fr

# 786. K-th Smallest Prime Fraction

https://leetcode.com/problems/k-th-smallest-prime-fraction/

priority_queue



for i
for j
i/j ==> priority_queue

if > K
priority_queue.poll()

Time Limit Exceeded

1 2 3 5 q = 1

binary search
while(true) 0.5 ( 0.75 , 0.25 )
0.5

1/3 1/5
2/5

p = 0
i
j = i +1

A[i] / A[j] < target

if p/q < A[i] / A[j]
update p = A[i], q =A[j]

p = 0

j =1

i

A[i] / A[j] < target

if p/q < A[i] / A[j]
update p = A[i], q =A[j]

代码魔术师
magiciendecode.fr

# 862. Shortest Subarray with Sum at Least K

https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/

| A | | 56 | -21 | 56 | 35 | -9 |
|---|---|---|---|---|---|---|
| sumsA | 0 | 56 | 35 | 91 | 126 | 117 |
| | 0 | 1 | 2 | 3 | 4 | 5 |

sums[end] - sums[start] >= K

sums[end] - K >= sums[start]

(sums,index)
sortedList: List<Pair<Int,Int>>

binary search the
last position that <= (sums[end]-K)

foreach 0 .. position
    result = minOf(
        result,
        nowIndex - second
    )

remove 0 untin position

(0,0)
(0,0) (56,1) -> 56-61 = -5
(0,0)(56,1) (35,2) -> 35-61 = -26
(0,0)(35,2)(56,1) (91,3) -> 91 -61 = 30 3
(35,2)(56,1)(91,3) (126,4) -> 126-61=65 2

(0,0) 56-0 >= K
(0,0)(56,1) 35

index_current > index_last
[current] < [last]
? - [current] >= K

(0,0)(35,2)

why we can't binary search
from 1 to A.size

how can we say a length is valid?

sums[i-length] - sums[i] >= K

if valid => right =mid
else -> left = mid

imagine that length = 1 is valid ..
eumm

代码魔术师
magiciendecode.fr

# 710. Random Pick with Blacklist

https://leetcode.com/problems/random-pick-with-blacklist/

0 1 2 3 4 5 6 7

hash[0] -> 5
hash[1] -> 2

0 1 2 3 4 5 6 7

hash[1] ->2

2 5
0 1 3 4 6 7 [0,5]

first element make random from[0,N-B.size] + index < B[index]

0 -> 0 1 (2) 3 4 5 -> 0 + 2 < 3
1 -> 1 2 3 4 (5) 6 -> 1 + 4 < 6
B 0 1 3 4 6 7

0 1
2

0
1 2

0 2
1

1
0 2

random + mid < B[mid]
-> right = mid
else -> left = mid

left/right + random
else -> random + right + 1

代码魔术师
magiciendecode.fr

# 793. Preimage Size of Factorial Zeroes Function

https://leetcode.com/problems/preimage-size-of-factorial-zeroes-function/

x = 1 ~ 4 K = 0
x = 5 ~ 9 K = 1
x = 10 ~ 14 K = 2
x = 15 ~ 19 K = 3
x = 20 ~ 24 K = 4
x = 25 ~ 29 K = 6

15 / 5 = 3 16 / 5 = 3

26 / 5 = 5 .. 1 5 / 5 = 1 .. 0

binary search from 0 to (K+1) * 5

if find a number that f(number) = K, return 5
else -> 0

K = 5 result = 0

```
private fun numberOfZero(long: Long): Int {
    var count = 0L
    var current = long
    while (current != 0L) {
        count += current / 5
        current /= 5
    }
    return count.toInt()
}
```

代码魔术师
magiciendecode.fr

# 878. Nth Magical Number

https://leetcode.com/problems/nth-magical-number/

```
// LCM, Least Common Multiple
private long lcm(long a, long b) {
    return a * b / gcd(a, b);
}
```

12 / 2 = 6  2  4  6  8  10  12
12 / 3 = 4  3  6  9  12

12 is N-th magical number for 2 and 3.  8

12 / 6 = 2  6  12

6 + 4 - 2 = 8

```
// GCD, Greatest Common Divisor
private long gcd(long a, long b) {
    if (a < b) {
        a = a ^ b;
        b = a ^ b;
        a = a ^ b;
    }
    while (a % b != 0) {
        final long remainder = a % b;
        if (b < remainder) {
            a = remainder;
        } else {
            a = b;
            b = remainder;
        }
    }
    return b;
}
```

代码魔术师
magiciendecode.fr

# 1044. Longest Duplicate Substring

https://leetcode.com/problems/longest-duplicate-substring/

a b a b a

b a b a
hahscode = b * 31(3) + a * 31(2) + b * 31(1) + a * 31(0) = long % (1000000007)

a b a = ( b a b a - b * 31(3) ) % (1000000007)

if(hashcode<0) ? + 1000000007

31*(target)
hash_map

currentCode

for index in string.indices
    if(index<target-1) continue
    if(index>=target) - if<0

    double check

    hashmap[currentCode] = index - target + 1

binar search from 1 to size -1

if(right)
    substring(right_result,right_result + right)

代码魔术师
magiciendecode.fr

# 1044. Longest Duplicate Substring

https://leetcode.com/problems/longest-duplicate-substring/

2 3 1 4

1 2 3 4

1 1 2 6

K = 9

ck = 8

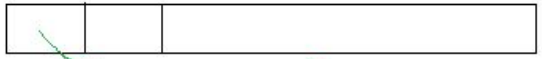
8/6 = 1
ck = ck % 6 = 2

2/2 = 1
ck = ck % 2 = 0

a,b,a,b,a

Trie (index = 0,depth = 0)

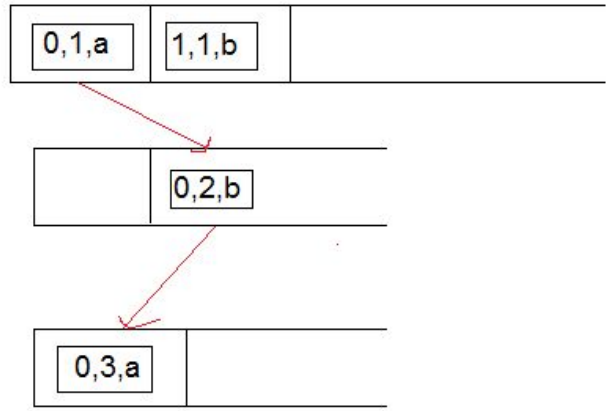
depth = 0
if(index+depth == S)



getIndex(0+0) a - 'a', Trie(0,1)

length = getIndex(1+0)

if( [length] null )
= Trie(1,0+1)
return depth
return (1,1)




代码魔术师
magiciendecode.fr

# 887. Super Egg Drop

https://leetcode.com/problems/super-egg-drop/

given floor ?

Time Limit Exceeded

```
DP[floor][egg] == minOf(
        for f in floor
                1 + maxOf(dp[f-1][egg-1],dp[floor-f][egg])
)
```

broken

not broken

K * N * log(N)

dp[time][egg], with eggs(K) how many times try can test the floor N

dp[time][0] = 0
dp[1][egg] = egg

dp[time][egg] = dp[time-1][egg-1] + dp[time-1][egg] + 1

if(dp >= N) return time

K * N

binary search, 2 .. floor
broken = dp[mid-1][egg-1]
notBroken = dp[floor-mid][egg]

broken < notBroken -> left = mid
else -> right = mid

binary search 1..N

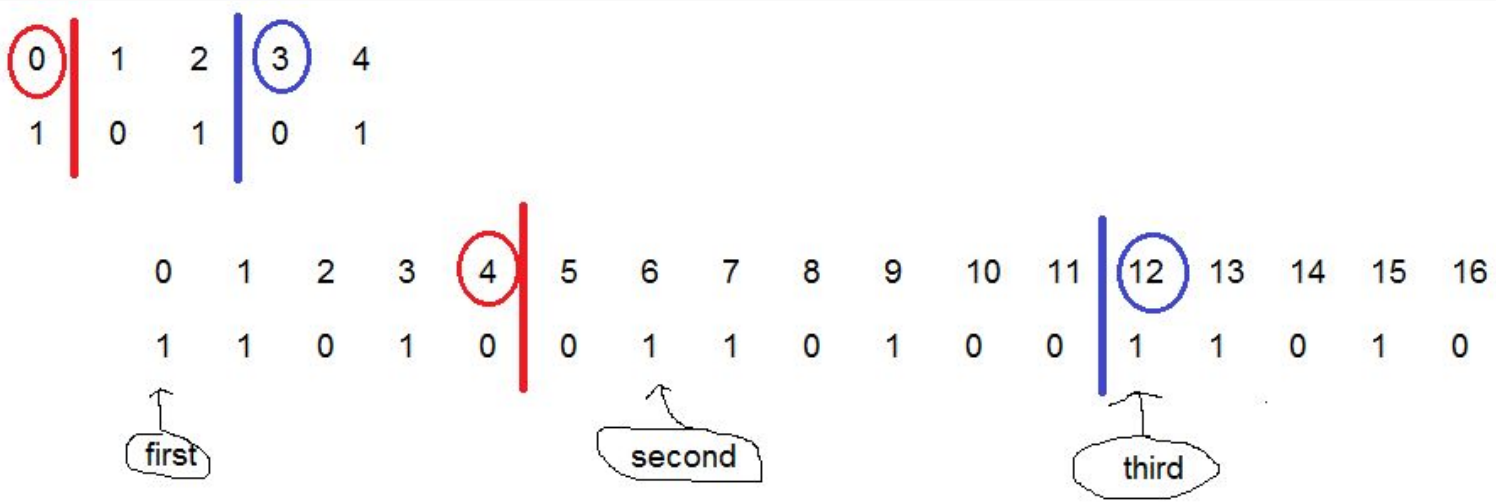
so hard, I recommend that we see the solution!

K * log(N)

代码魔术师
magiciendecode.fr

# 927. Three Equal Parts

https://leetcode.com/problems/three-equal-parts/

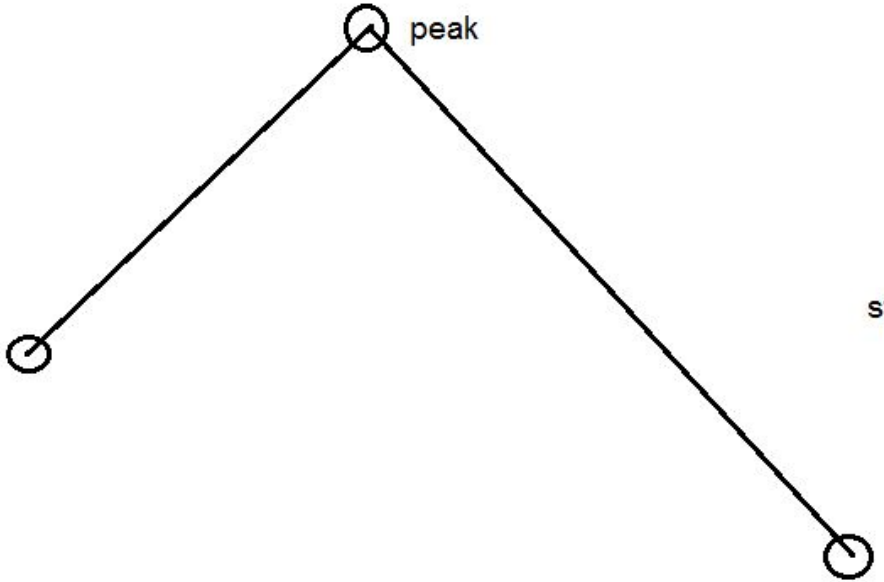



代码魔术师
magiciendecode.fr

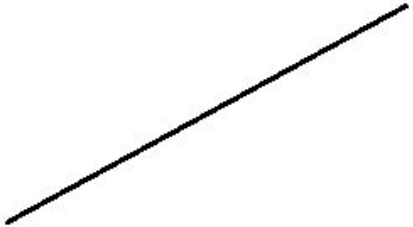
# 1095. Find in Mountain Array

https://leetcode.com/problems/find-in-mountain-array/

step1: binary search find the peak




step2: if target is in [0] to [peak], binary search



step3: if target is in [peak] to [length-1], binary search



代码魔术师
magiciendecode.fr

# 1157. Online Majority Element In Subarray

https://leetcode.com/problems/online-majority-element-in-subarray/

摩尔投票法

Boyer-Moore Majority Vote Algorithm

```
var major = -1
var count = 0
for (index in left..right) {
    if (count == 0) {
        major = arr[index]
        ++count
    } else {
        count += if (arr[index] == major) 1 else -1
    }
}
count = 0
for (index in left..right) {
    if (arr[index] == major) {
        ++count
    }
}
return if (count >= threshold) major else -1
```

we need a list of index,
for example, [1,1,2,2,1,1]
the list of index for 1 is [0,1,4,5]

if we want to find majority element from index 1 to 5,

just find the index in list then the number of 1 is 3.

we can compare it with threshold.

so how many time ?

代码魔术师
magiciendecode.fr

# 1482. Minimum Number of Days to Make m Bouquets

https://leetcode.com/problems/minimum-number-of-days-to-make-m-bouquets/

1 10 3 10 2
m = 3
k = 1

when target = 3

the array could be
O _ O _ O

k = 1
so total is 3 >= m = 3

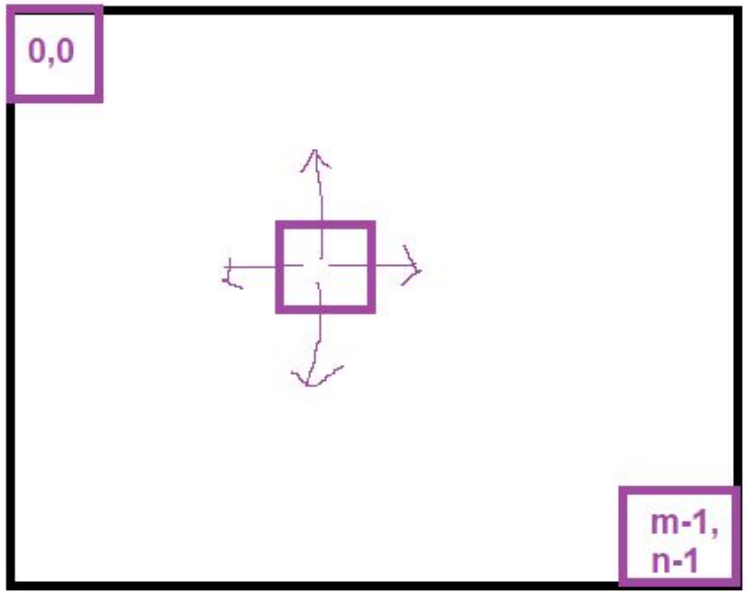
binary search from min and max of array,
    if( isvalid )
        right = mid
    else
        left = mid

if left is valid -> left
if right is valid -> right
else -> -1

代码魔术师
magiciendecode.fr

# 1631. Path With Minimum Effort

https://leetcode.com/problems/path-with-minimum-effort/




effort [ 0 ~~ 1 000 000 ]

4 directions:
- inBound
- not visited
- diff <= effort

Binary Search
+
Breadth First Search

代码魔术师
magiciendecode.fr

# 870. Advantage Shuffle

https://leetcode.com/problems/advantage-shuffle/

A: [ 12, 24, 8, 32 ]
B: [ 13, 25, 32, 11 ]

sortedA: [ 8, 12, 24, 32 ]

1. sorted A
2. find the first element (return index) that greater than target in B

if index == -1, not found, take and remove element at index 0
if index !=-1, found, take and remove element at index

B [ 13 ], sortedA --> 24
sortedA: [ 8, 12, 32 ]

B [ 25 ], sortedA --> 32
sortedA: [ 8, 12 ]

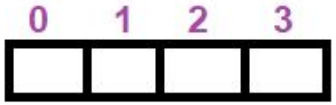
B [ 32 ], not found, take sortedA --> 8
sortedA: [ 12 ]

二分查找复习

代码魔术师
magiciendecode.fr

# 295. Find Median from Data Stream

https://leetcode.com/problems/find-median-from-data-stream/



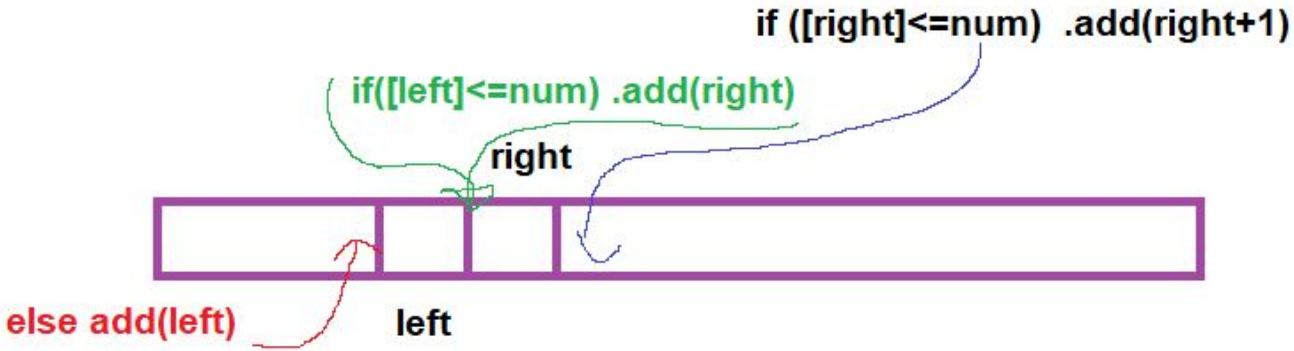

size % 2 == 0

( (size/2) + (size/2 - 1) ) / 2.0



size % 2 == 1

(size/2)




代码魔术师
magiciendecode.fr